Chapter V: Mendel’s law

First, a little more genetics terminology.

Then…



Mendel's Laws

1. Law of Dominance

2. Law of Segregation

3. Law of Independent Assortment

Mendel's Law of Dominance

- For each character, an organism inherits 2 alleles, 1 from each parent
- These alleles may be the same or different
- An organism that has 2 identical alleles for a gene is said to be **homozygous**
- An organism that has 2 different alleles for a gene is said to be **heterozygous**

Mendel's Law of Dominance

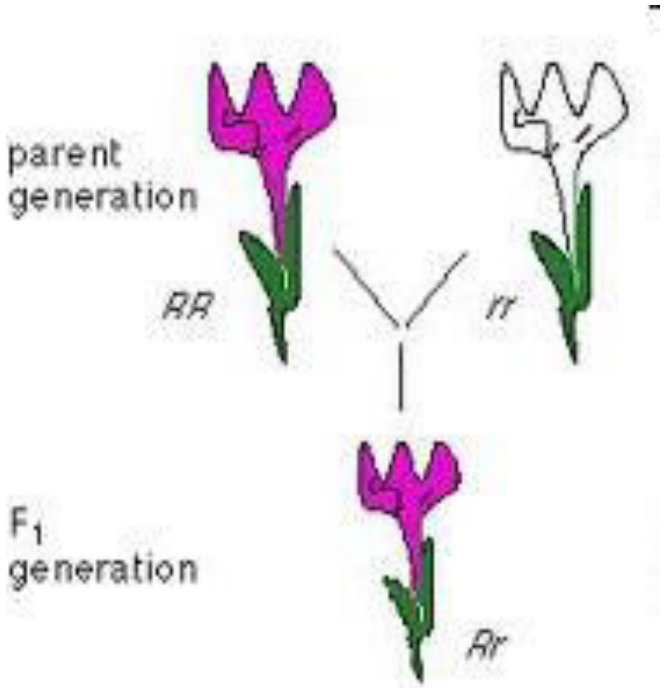
- If the 2 alleles of an inherited pair differ, then one allele will determine the organism's appearance over the other, and is called the **dominant allele**
- The other allele has no noticeable effect on the organism's appearance and is called the **recessive allele**
  - We use upper and lower case letters to describe the dominant and recessive alleles, respectively

Law of Dominance

- Every gene has two alleles that can code for a trait.
- One allele is dominant, meaning it will always show.
- One allele is recessive, meaning it will be masked by the presence of the dominant allele.
- Hybrids will always show the dominant phenotype.

Ex: PP = purple pp = white

<u>Pp = purple</u>

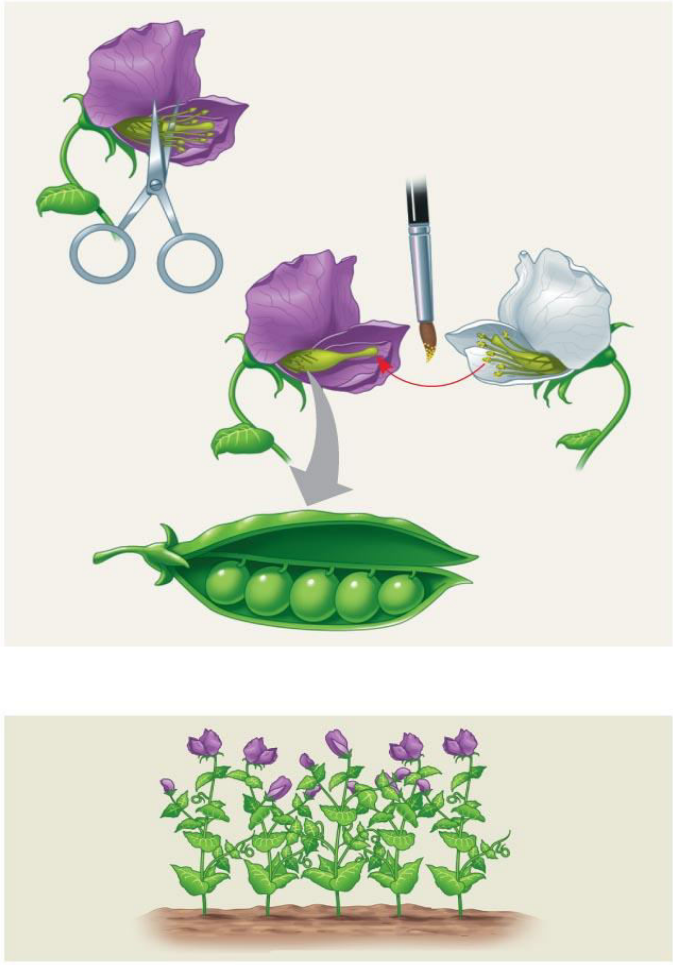


		
Flower color	Purple	White
		
Flower position	Axial	Terminal
		
Seed color	Yellow	Green
		
Seed shape	Round	Wrinkled
		
Pod shape	Inflated	Constricted
		
Pod color	Green	Yellow
		
Stem length	Tall	Dwarf

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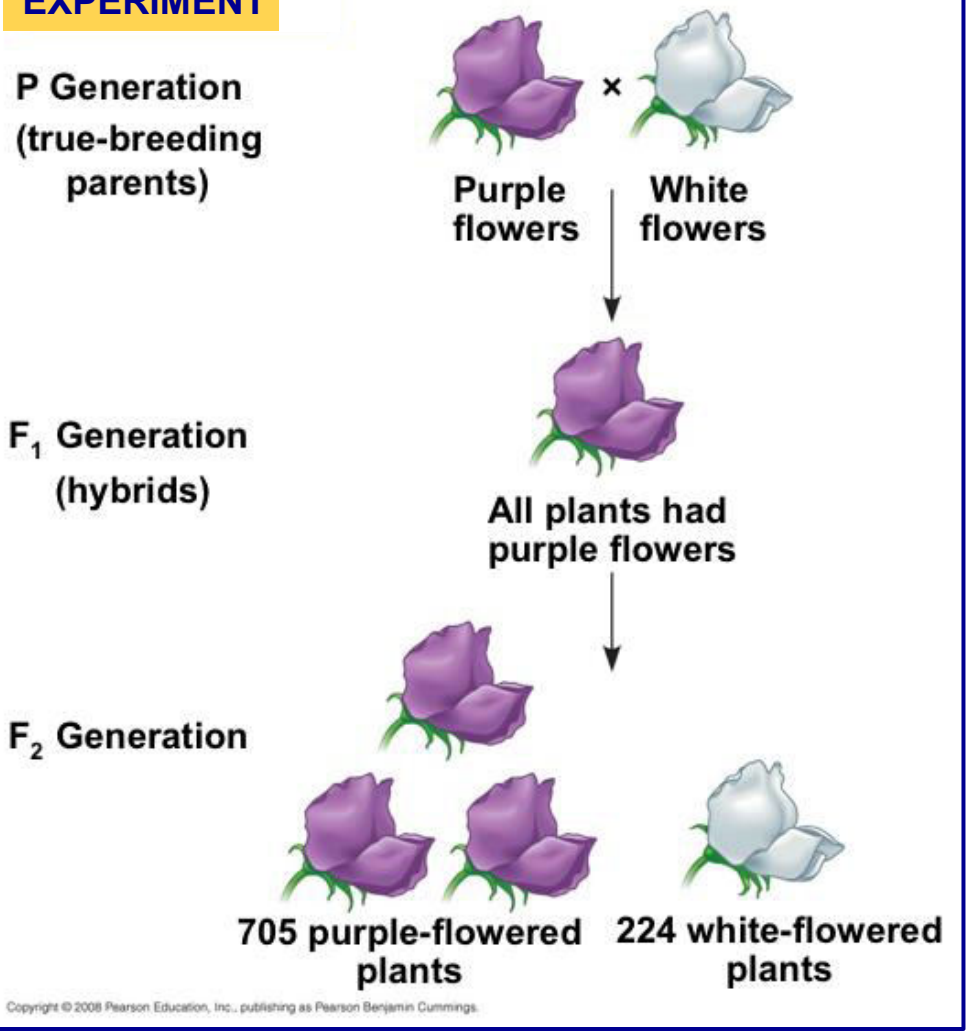
Mendelian Genetics …

TECHNIQUE



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EXPERIMENT



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Mendel crossed two <u>pure strains</u> of pea plants with purple and white flowers and discovered that the <u>first filial generation</u> were all purple

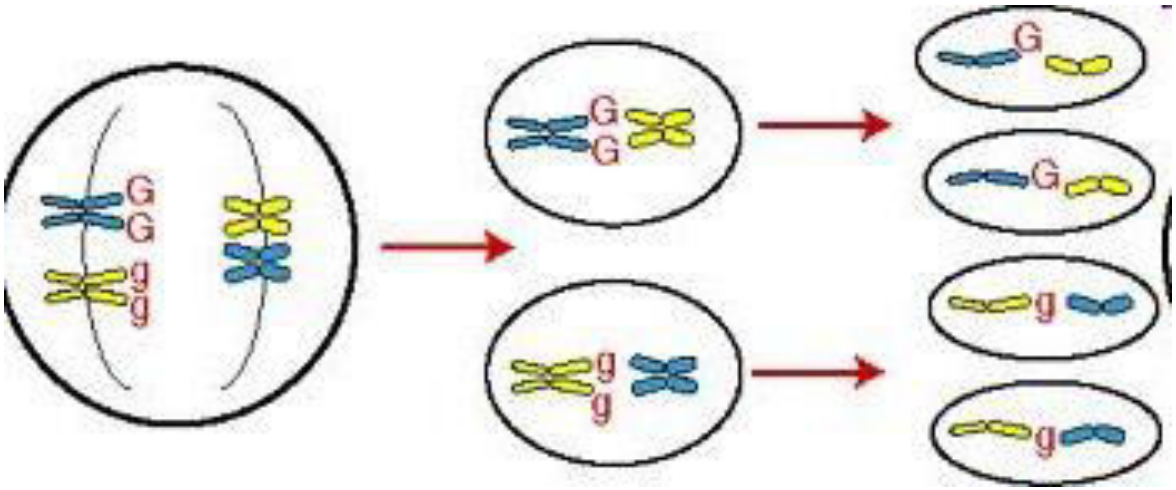
Law of Segregation

- Mendel arrived at this conclusion by performing monohybrid crosses.
- These were cross-pollination experiments with pea plants that differed in one trait, for example height of pea plant.
- The alleles for a trait separate when gametes are formed. These alleles can then randomly united at fertilization.

Dominant and Recessive
(T = Tall & t = short
Cross: Tt x Tt

	T	t
T	TT	Tt
t	Tt	tt

Genotypic ratio: 1 : 2 : 1 (TT=25% Tt=50% tt=25%)
Phenotypic ratio: 3 : 1 (Tall=75% Short=25%)



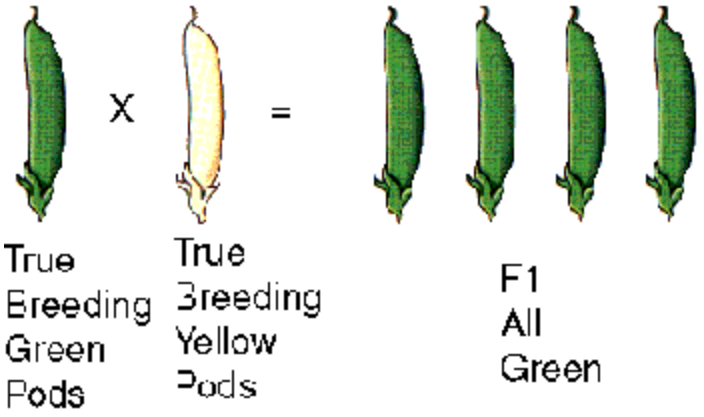
Anaphase I of Meiosis **Anaphase II of Meiosis**

Experiments with Pea Plants

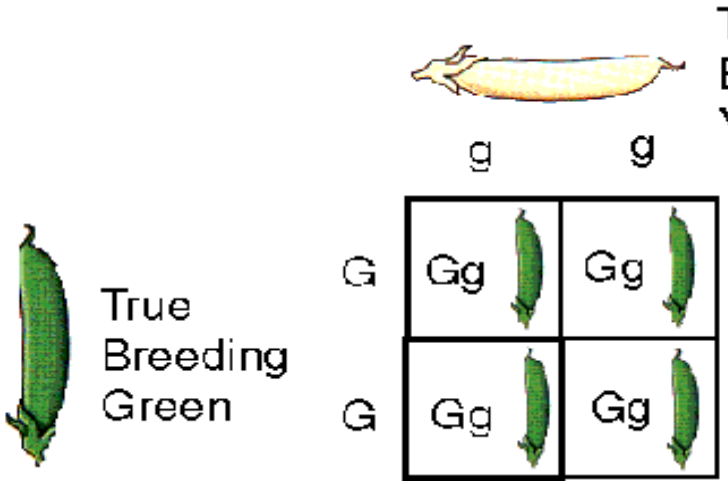
- Seed coat colour (gray or white)
- Seed shape (round or wrinkled)
- Seed colour (yellow or green)
- Pod colour (green or yellow)
- Flower position (axial or terminal)
- Pod shape (inflated or constricted)
- Stem length (tall or dwarf)



Cross-Pollination of Purebread Plants



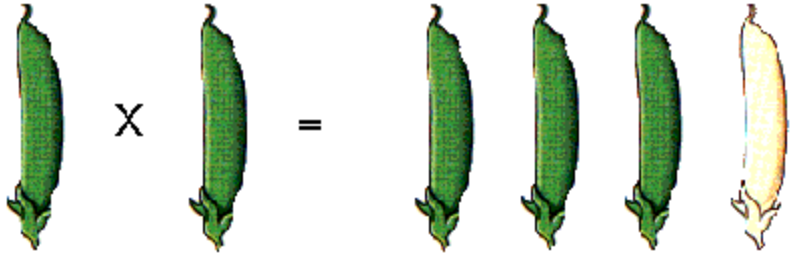
- cross-pollination between true breeding green and yellow pods
- all F1 green



F1 Generation

Gg = heterozygous

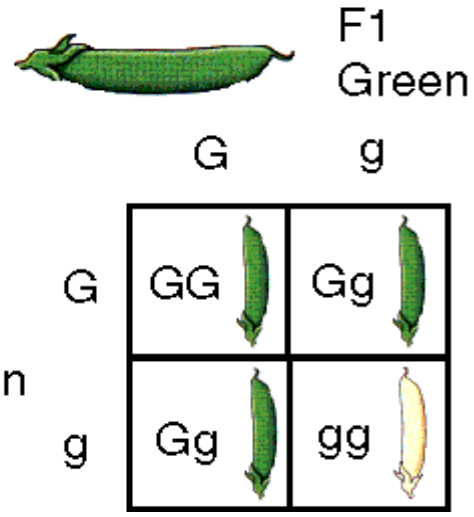
F2 Generation



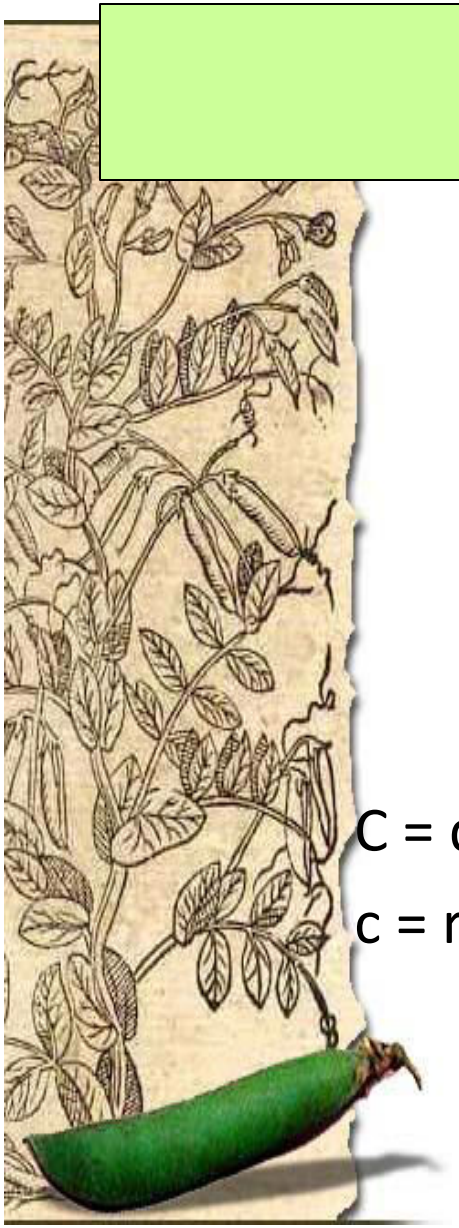
- self-pollination of green F1 plants
- ¾ in F2 green, ¼ yellow
- 3 : 1 ratio in pod colour in F2

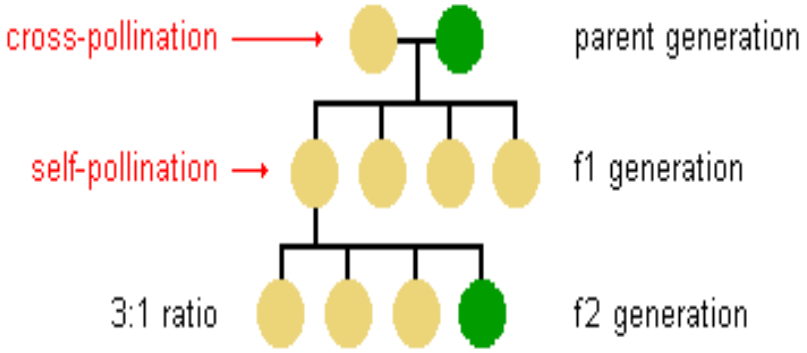
G = dominant = green
g = recessive = yellow

GG, gg = homozygous



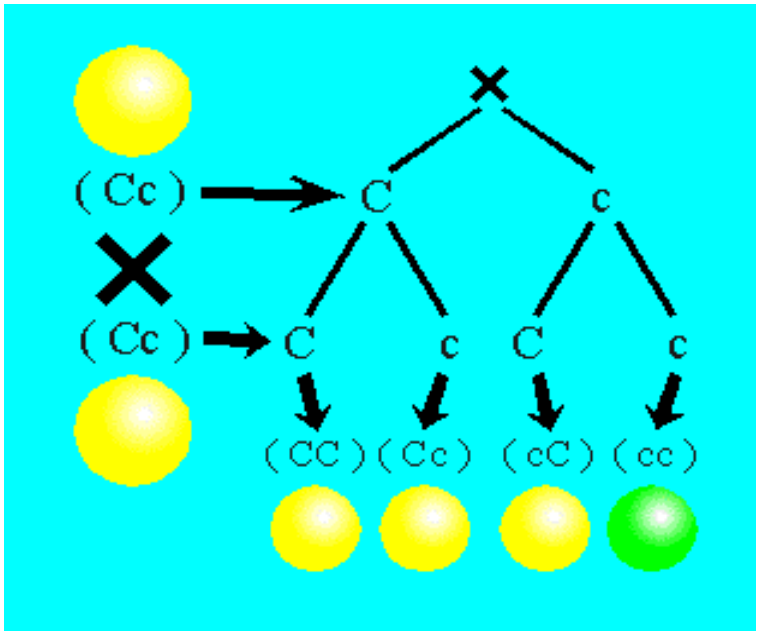
Seed Colour





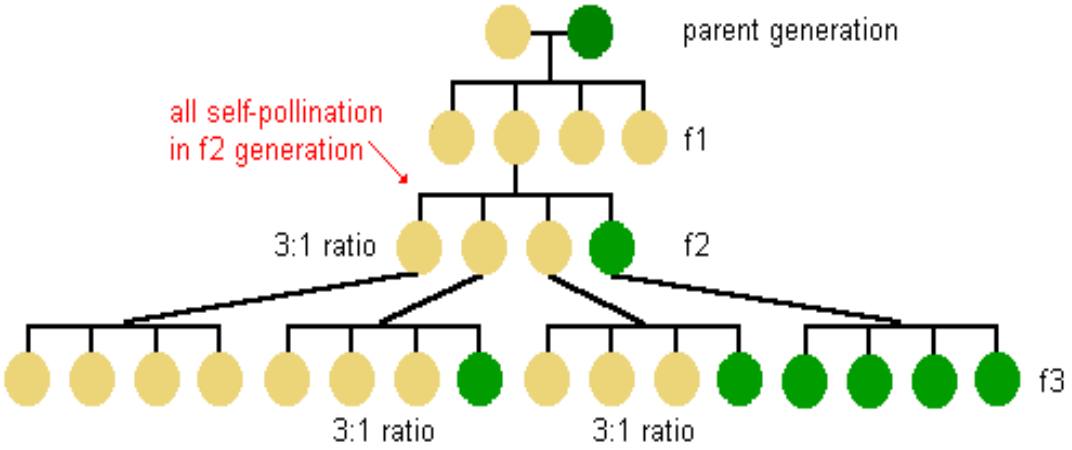
C = dominant = yellow

c = recessive = green

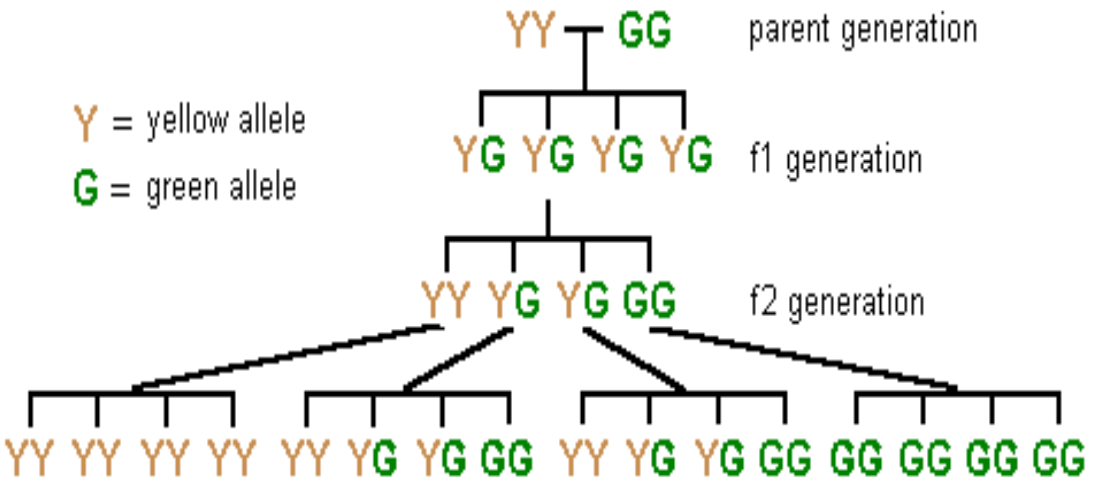


Inheritance of Pea Colour

phenotype:



genotype:



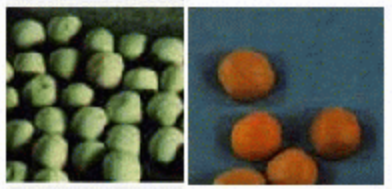
Mendelian Genetics – 3:1 ratios …

Traits must come in discrete units – *now called genes*

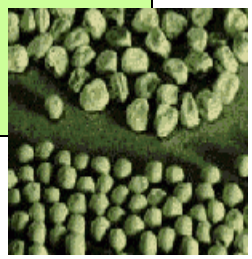
Table 14.1 The Results of Mendel's F_1 Crosses for Seven Characters in Pea Plants

Character	Dominant Trait	x	Recessive Trait	F_2 Generation Dominant:Recessive	Ratio
Flower color	Purple 	×	White 	705:224	3.15:1
Flower position	Axial 	×	Terminal 	651:207	3.14:1
Seed color	Yellow 	×	Green 	6,022:2,001	3.01:1
Seed shape	Round 	×	Wrinkled 	5,474:1,850	2.96:1
Pod shape	Inflated 	×	Constricted 	882:299	2.95:1
Pod color	Green 	×	Yellow 	428:152	2.82:1
Stem length	Tall 	×	Dwarf 	787:277	2.84:1

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Mendel‘s Generalization



1. Alternative versions of genes account for variations in inherited characters

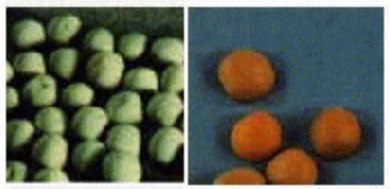
   - concept of alleles (G=green, g=yellow)

2. For each character, an organism inherits two genes, one from each parent

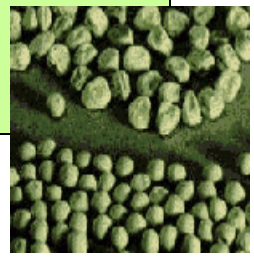
   - two gametes form somatic cells

   - one allele comes from the mother,

     one from the father



Mendel‘s Generalization



3. If the two alleles differ, then:

   - dominant allele is fully expressed in the organism's appearance (phenotype)

   - recessive allele has no noticeable effect on the organism's appearance (genotype)

4. The two genes for each character segregate during gamete production

   - ensures variation

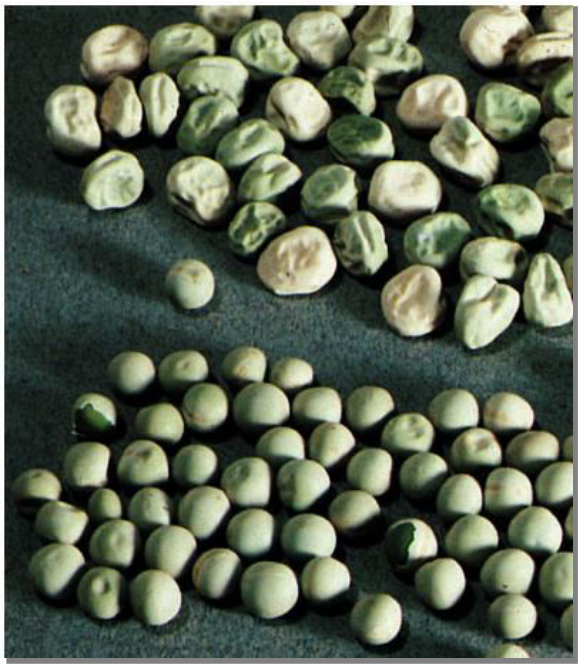


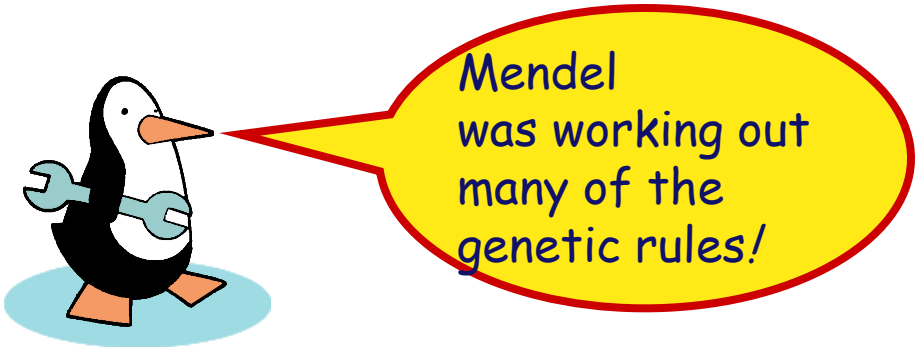
Law of Segregation

- the pair of alleles of each parent separate and only one allele passes from each parent on to an offspring
- which allele in a parent's pair of alleles is inherited is a matter of chance
- segregation of alleles occurs during the process of gamete formation (meiosis)
- randomly unite at fertilization

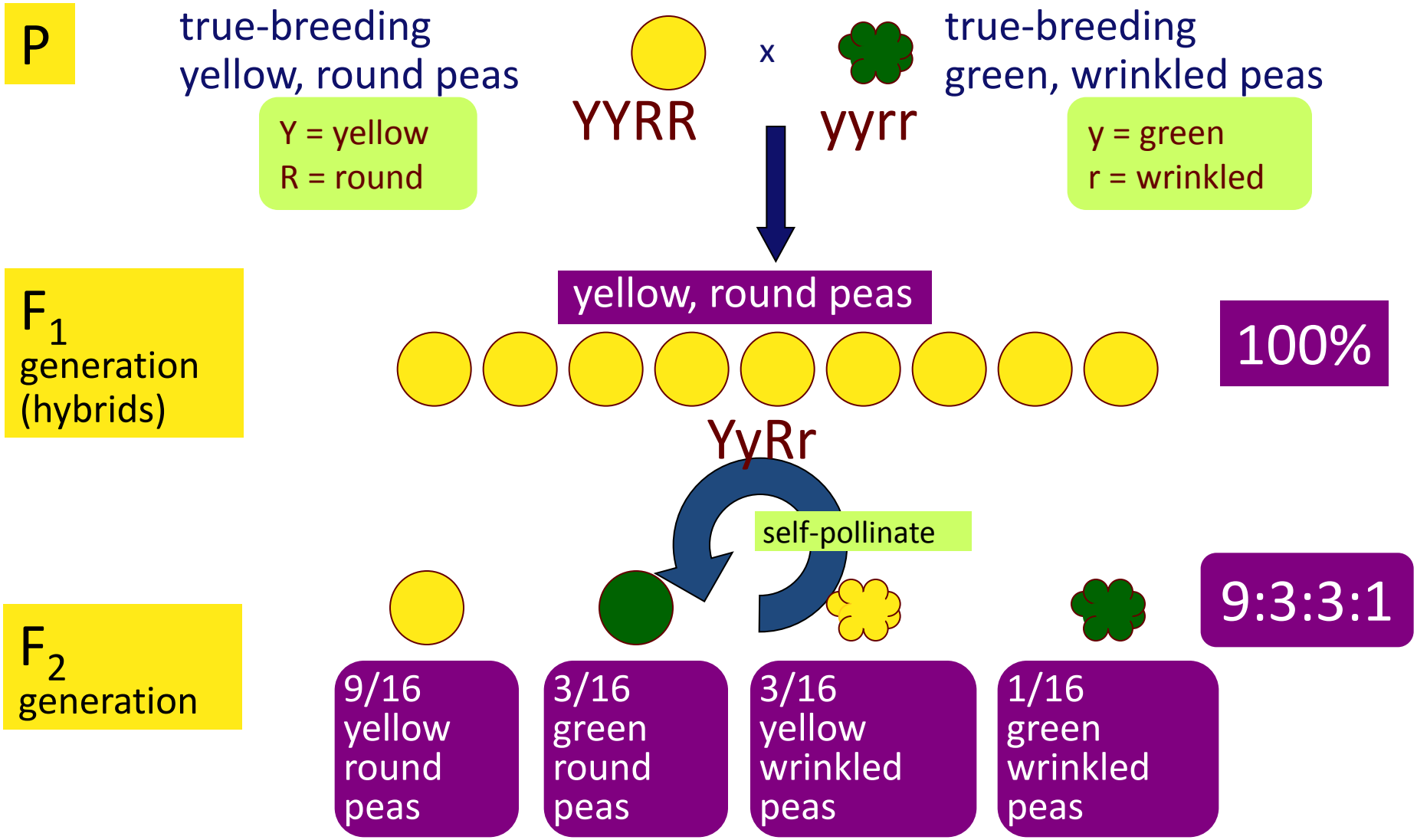
Dihybrid cross

- Other of Mendel's experiments followed the **inheritance of 2 different characters**
  - seed color and seed shape
  - dihybrid crosses



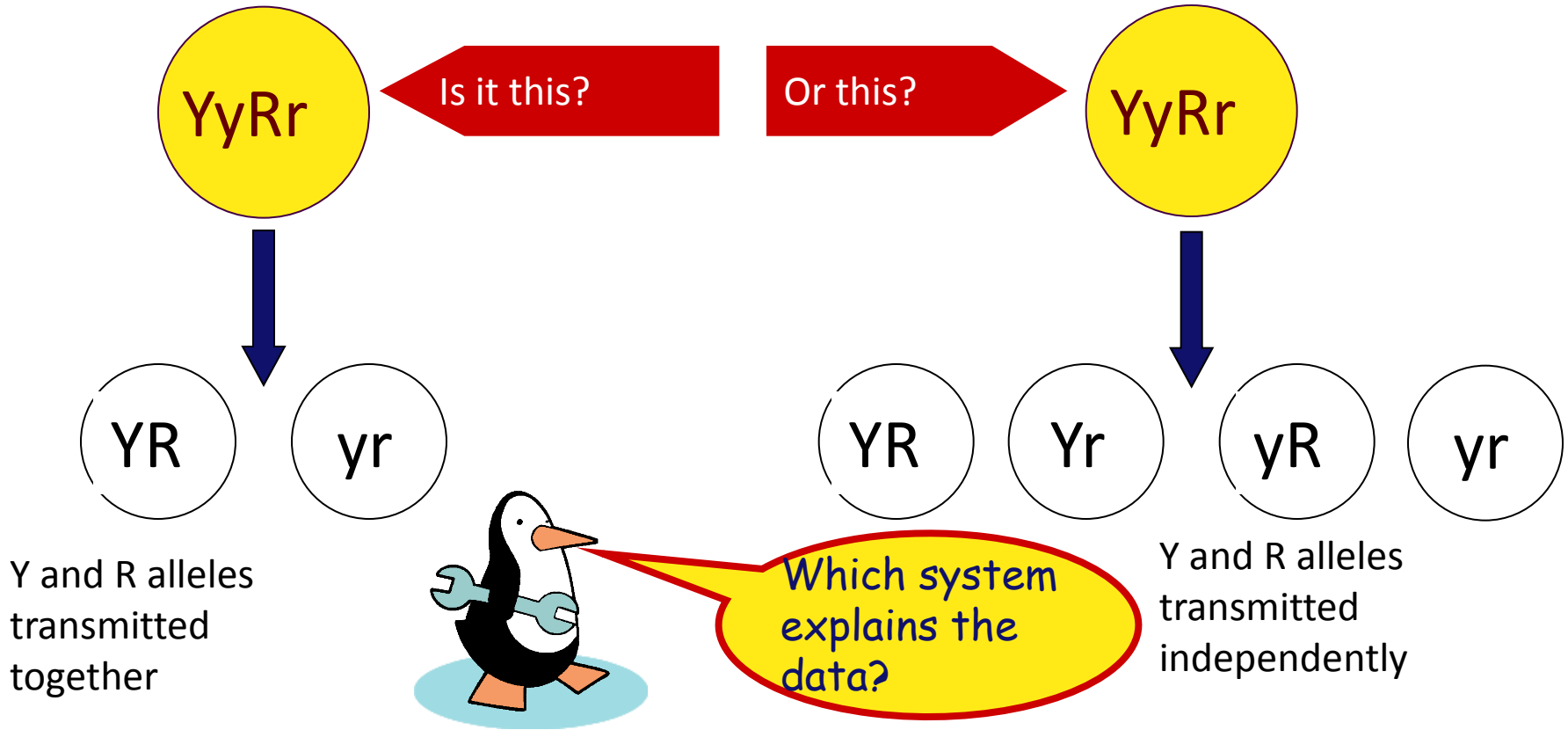


Dihybrid cross

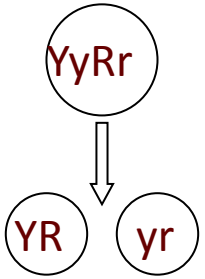


What's going on here?

- If alleles (different characteristics) are on different chromosomes…
  - Are alleles transmitted from parents to offspring together or independently?
  - Will the Y and R alleles always stay together, generation after generation?



Is this the way it works?

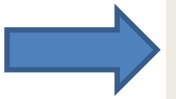


YyRr x YyRr



	YR	yr
YR	YYRR	YyRr
yr	YyRr	yyrr

What was actually observed





9/16 yellow round



3/16 green round

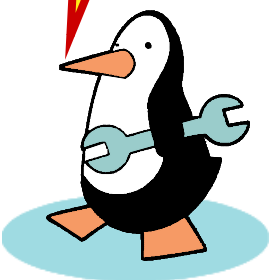


3/16 yellow wrinkled

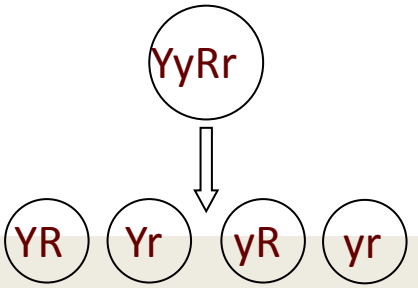


1/16 green wrinkled





Dihybrid cross



YyRr x YyRr

	YR	Yr	yR	yr
YR	YYRR	YYRr	YyRR	YyRr
Yr	YYRr	YYrr	YyRr	Yyrr
yR	YyRR	YyRr	yyRR	yyRr
yr	YyRr	Yyrr	yyRr	yyrr

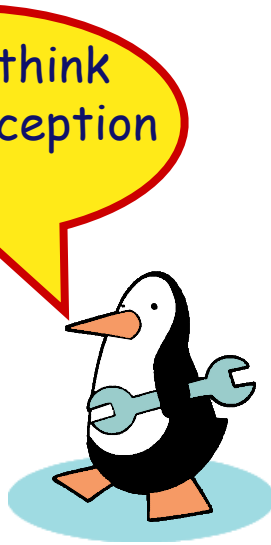
9/16 yellow round

3/16 green round

3/16 yellow wrinkled

1/16 green wrinkled

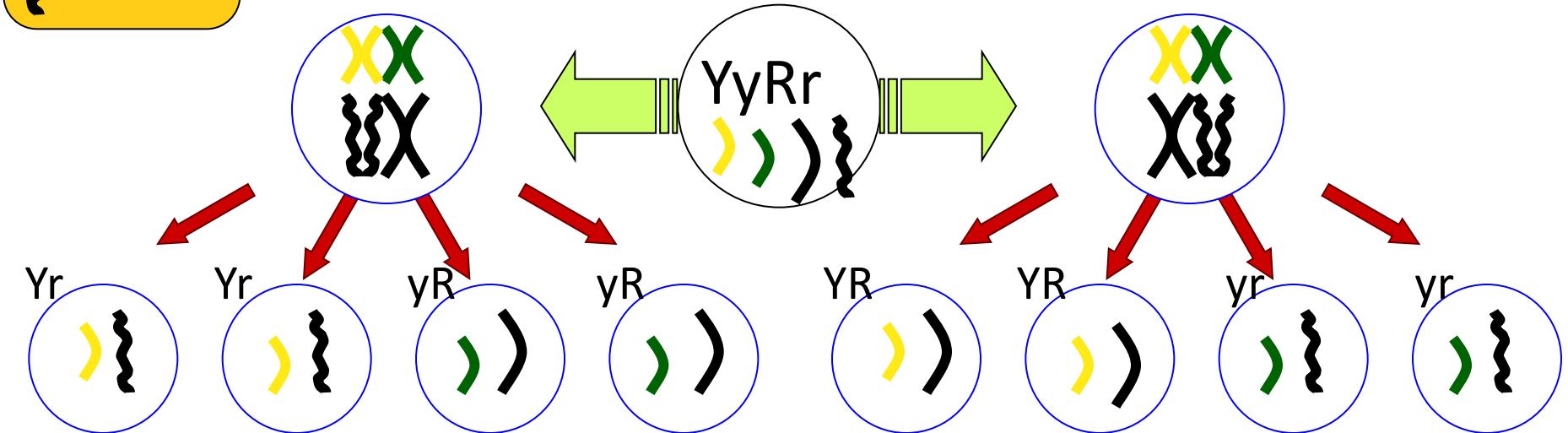
Mendel's 2nd law of her



- Law of independent assortment
  - Each pair of alleles (different characteristics) segregate independently
    - ***non-homologous*** chromosomes align independently
    - only true if alleles (different characteristics) are on separate chromosomes (e.g. gene for colour on separate chromosome as gene for texture)

yellow
green

round
wrinkled

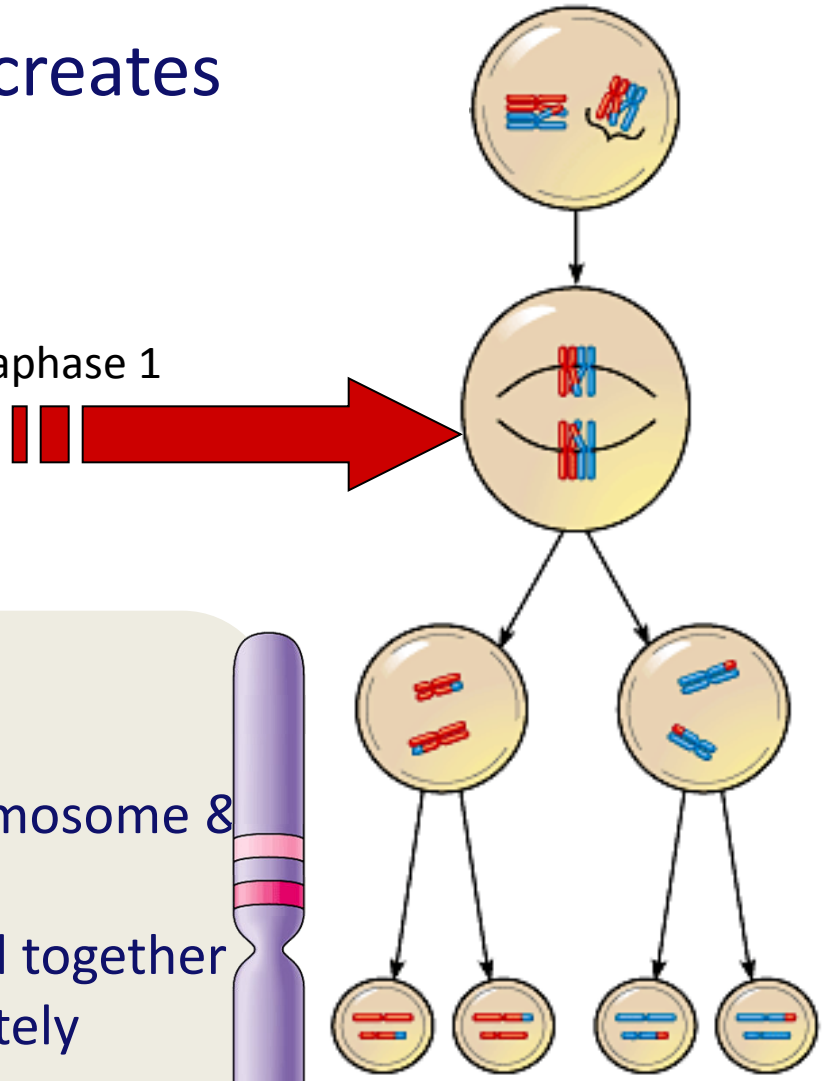


Law of Independent Assortment

- Which stage of meiosis creates the law of <u>independent assortment</u>?

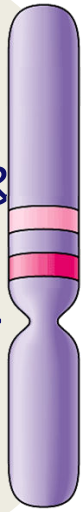


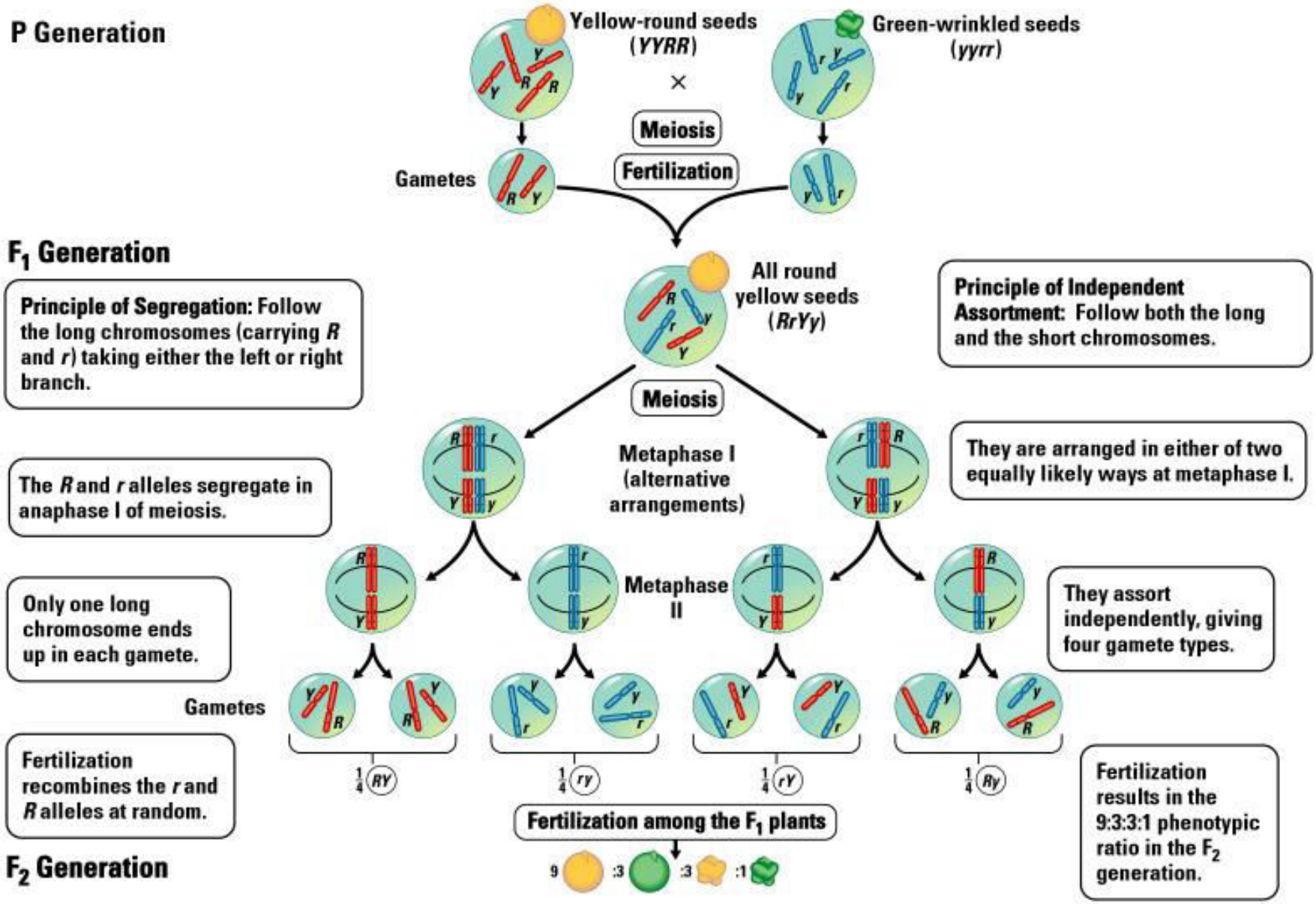
Metaphase 1



EXCEPTION

- If genes are on same chromosome & close together
  - will usually be inherited together
  - rarely crossover separately
  - "<u>linked</u>"





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Exercise:

In pea plants, the green colour allele (G) is dominant over the yellow colour allele (g) for seed colour and tall (T) is the dominant plant height. A pea plant that is heterozygous for both seed colour and height is crossed with a plant that is homozygous recessive for both.

Determine the genotypic ratio for their offspring.

Solution

1:1:1:1

GgTt x ggtt

	GT	Gt	gT	gt
gt	GgTt	Ggtt	ggTt	ggtt
gt	GgTt	Ggtt	ggTt	ggtt
gt	GgTt	Ggtt	ggTt	ggtt
gt	GgTt	Ggtt	ggTt	ggtt

4/16 = 1/4
green
tall

4/16 = 1/4
green
short

4/16 = 1/4
yellow
tall

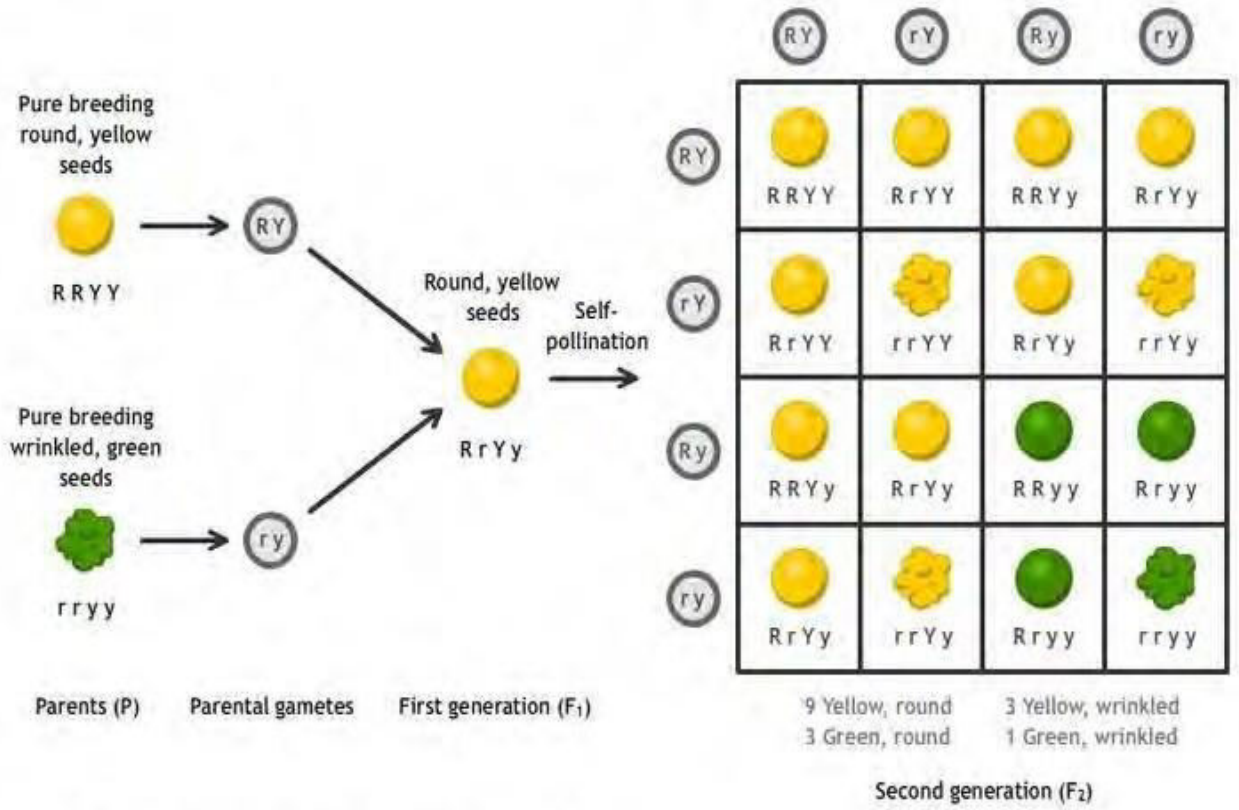
4/16 = 1/4
yellow
short

- What is the probability of getting an offspring that are heterozygous for both characters?

  A: They must be GgTt

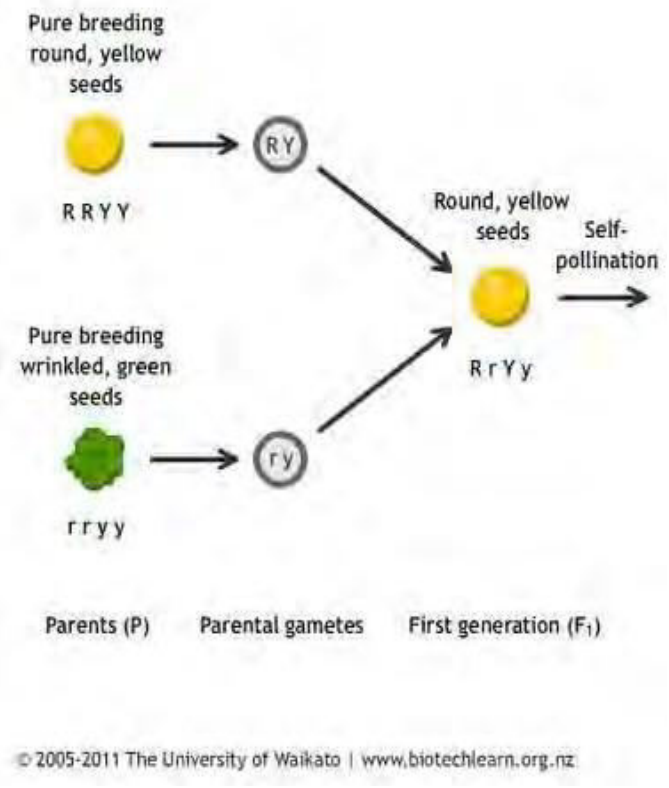
  4/16 = ¼ or 25%

If the transmission is independent:



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Law of independent assortment



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Mendel's results:

315	ratio = 9
108	ratio = 3
101	ratio = 3
32	ratio = 1

Mendel's conclusions

- Genes are distinct entities that remain unchanged during crosses
- Each plant has two alleles of a gene
- Alleles segregated into gametes in equal proportions, each gamete got only one allele
- During gamete fusion, the number of alleles was restored to two

Summary of Mendel's Principles

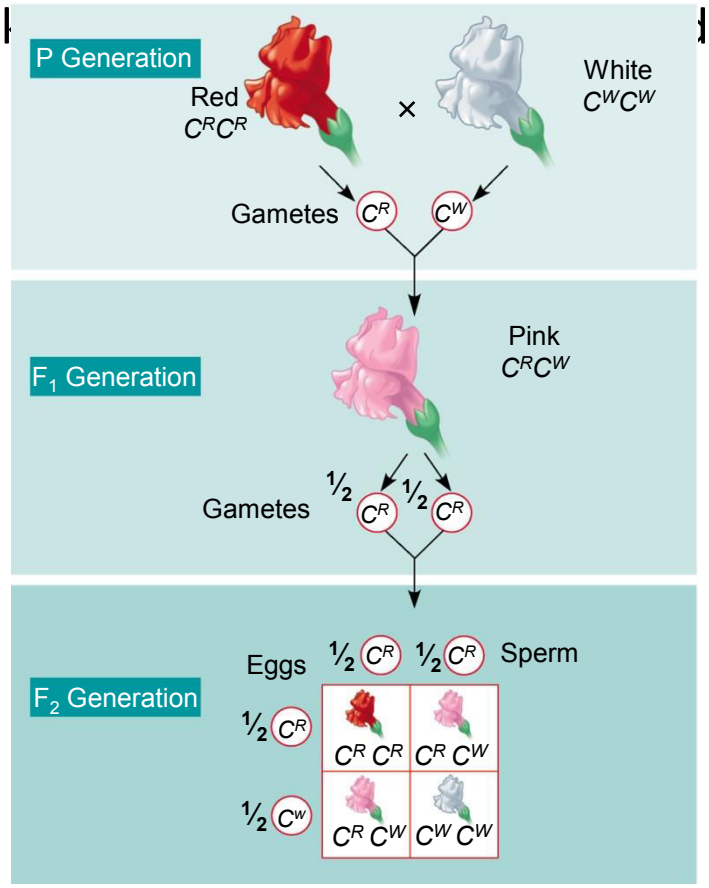
- Mendel's Principle of Uniformity in F1:
  - F1 offspring of a monohybrid cross of true-breeding strains resemble only one of the parents.
  - Why? Smooth seeds (allele S) are completely dominant to wrinkled seeds (alleles).
- Mendel's Law of Segregation:
  - Recessive characters masked in the F1 progeny of two true-breeding strains, reappear in a specific proportion of the F2 progeny.
  - Two members of a gene pair segregate (separate) from each other during the formation of gametes.
- Mendel's Law of Independent Assortment:
  - Alleles for different traits assort independently of one another.
  - Genes **on different chromosomes** behave independently in gamete production.

Exceptions To Mendel's Original Principles

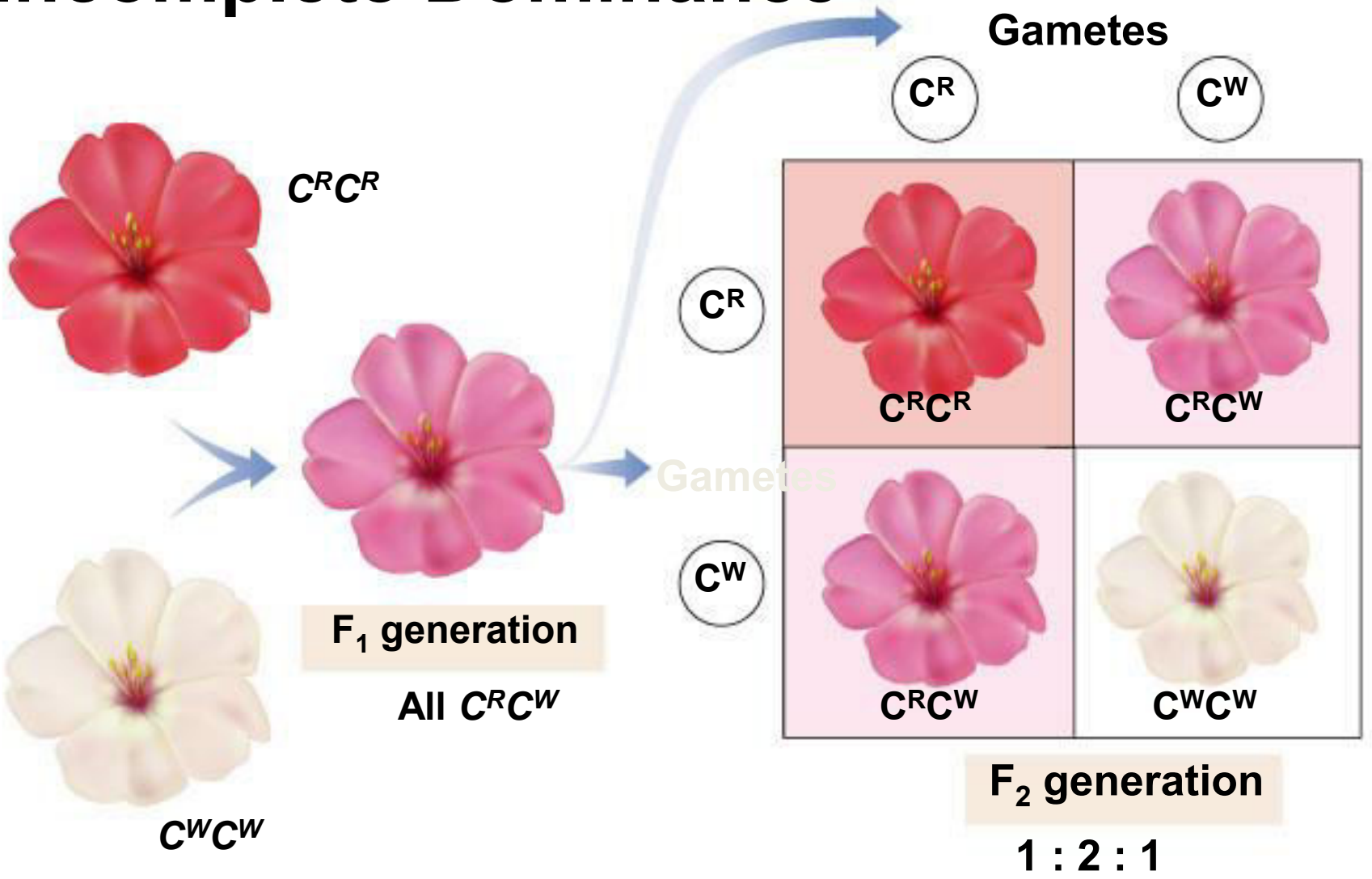
- Incomplete dominance
- Codominance
- Multiple alleles
- Polygenic traits
- Epistasis
- Pleiotropy
- Environmental effects on gene expression
- Linkage
- Sex linkage

Incomplete dominance

- Neither allele is dominant and heterozygous individuals have an intermediate phenotype
- For example, in Japanese “Four o’cloc[illegible] one white allele have pink flowers:



Incomplete Dominance

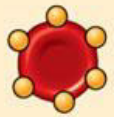


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Codominance

- Neither allele is dominant and both alleles are expressed in heterozygous individuals
- Example ABO blood types

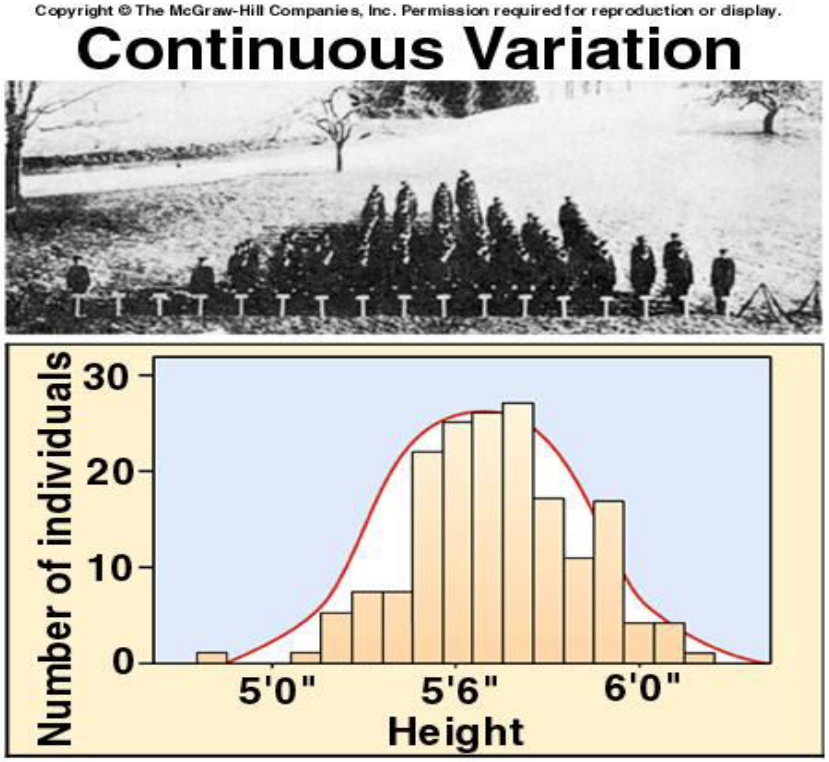
Table 14.2 Determination of ABO Blood Group by Multiple Alleles

Genotype	Phenotype (Blood Group)	Red Blood Cells
I^AI^A or I^Ai	A	
I^BI^B or I^Bi	B	
I^AI^B	AB	
ii	O	

Polygenic Traits

- Most traits are not controlled by a single gene locus, but by the combined interaction of many gene loci. These are called polygenic traits.
- Polygenic traits often show continuous variation, rather then a few discrete forms:

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Epistasis

- Type of polygenic inheritance where the alleles at one gene locus can hide or prevent the expression of alleles at a second gene locus.
- Labrador retrievers one gene locus affects coat color by controlling how densely the pigment eumelanin is deposited in the fur.
- A dominant allele (B) produces a black coat while the recessive allele (b) produces a brown coat
- However, a second gene locus controls whether any eumelanin at all is deposited in the fur. Dogs that are homozygous recessive at this locus (ee) will have yellow fur no matter which alleles are at the first locus:

Epistasis

E_
Dark pigment in fur

ee
No dark pigment in fur

eebb — **Yellow fur**

eeB_ — **Yellow fur**

E_bb — **Brown fur**

E_B_ — **Black fur**









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Pleiotropy

- This is when a single gene locus affects more than one trait.
- For example, in Labrador retrievers the gene locus that controls how dark the pigment in the hair will be also affects the color of the nose, lips, and eye rims.

Environmental Effects on Gene Expression

- The phenotype of an organism depends not only on which genes it has (genotype), but also on the environment under which it develops.

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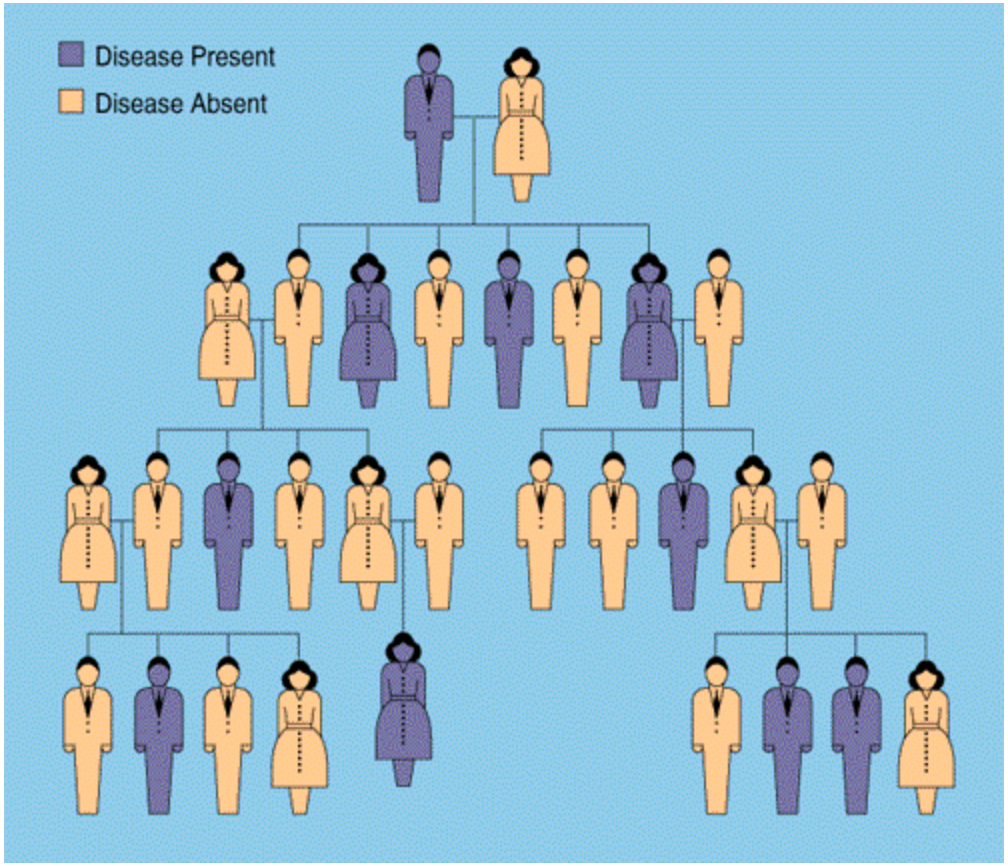
Arctic Fox in Winter



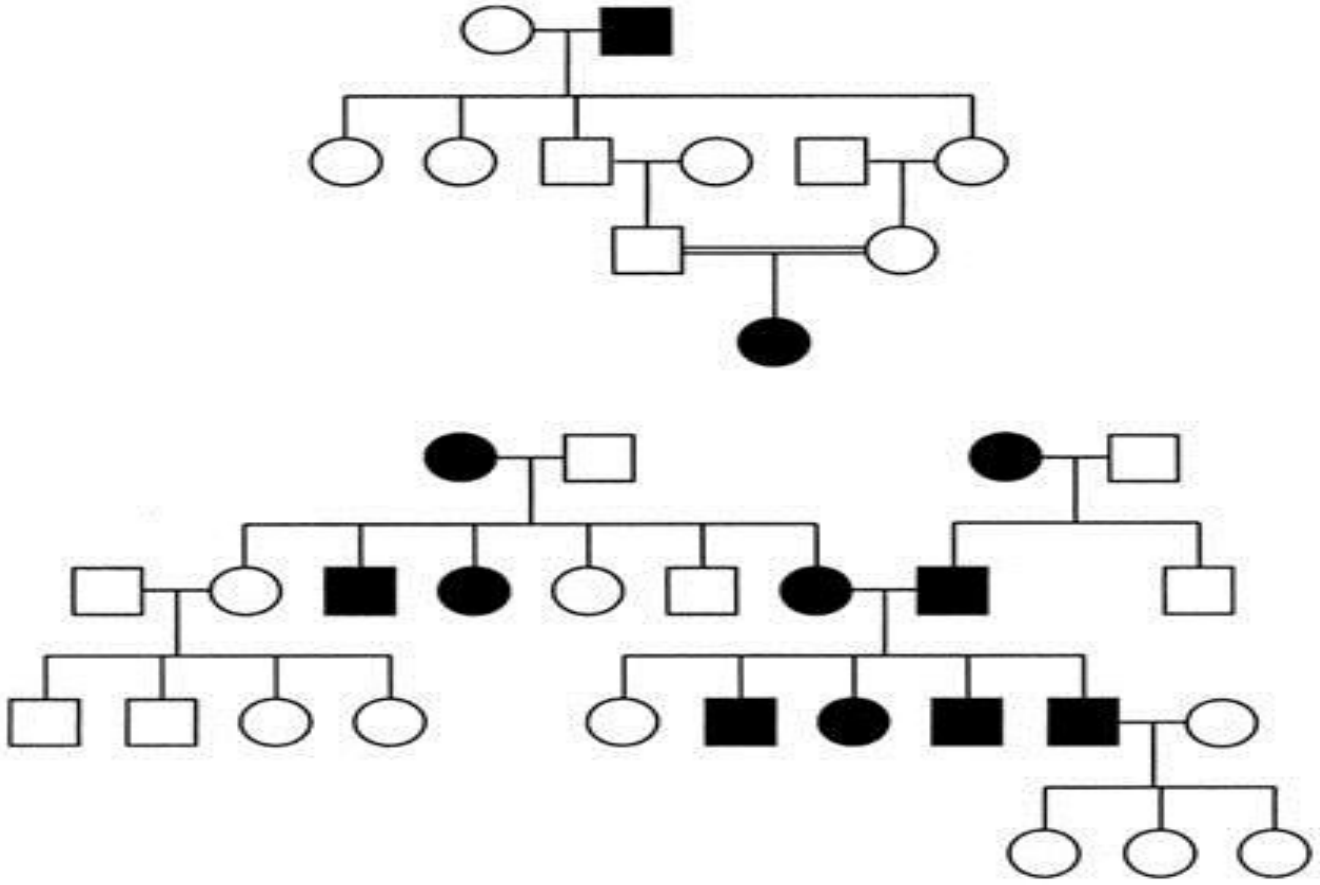
Arctic Fox in Summer

- Although scientists agree that phenotype depends on a complex interaction between genotype and environment, there is a lot of debate and controversy about the relative importance of these 2 factors, particularly for complex human traits.

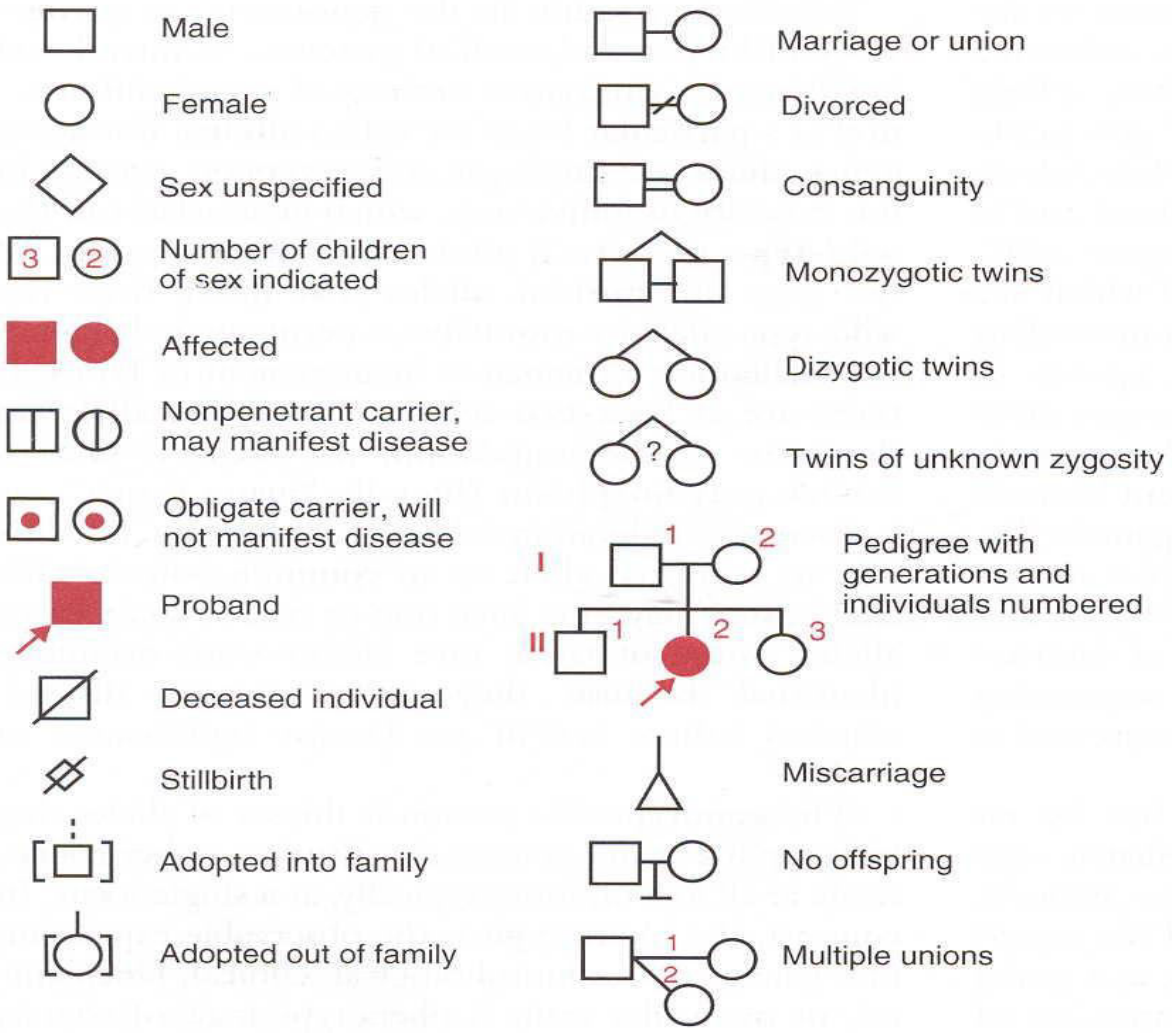
Mendelian human diseases



Pedigree



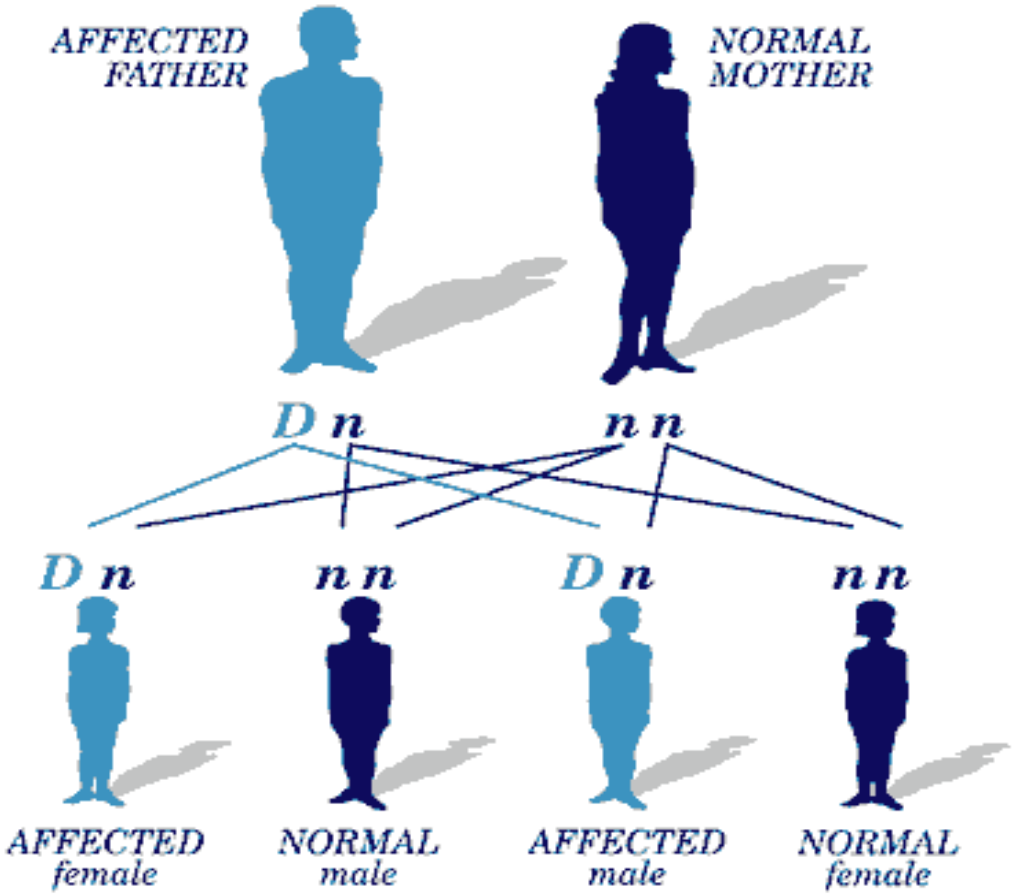
Pedigree symbols



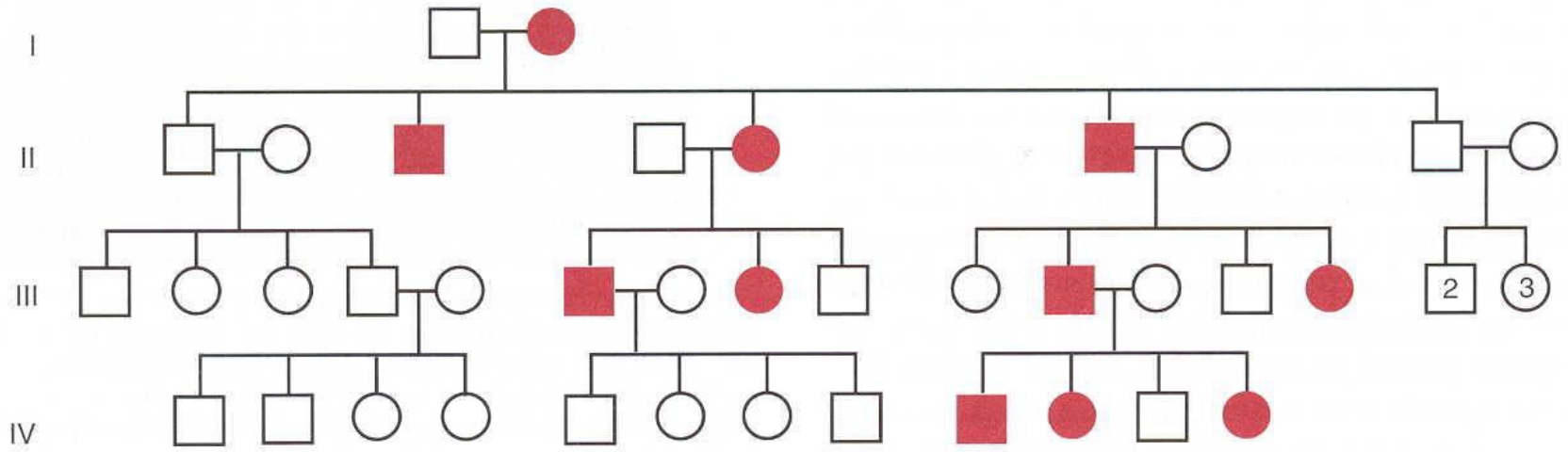
Genetic disorders with classical Mendelian inheritance

	Dominant	**Recessive**
Autosomal	Autosomal dominant	Autosomal recessive
X-linked	X-linked dominant	X-linked recessive

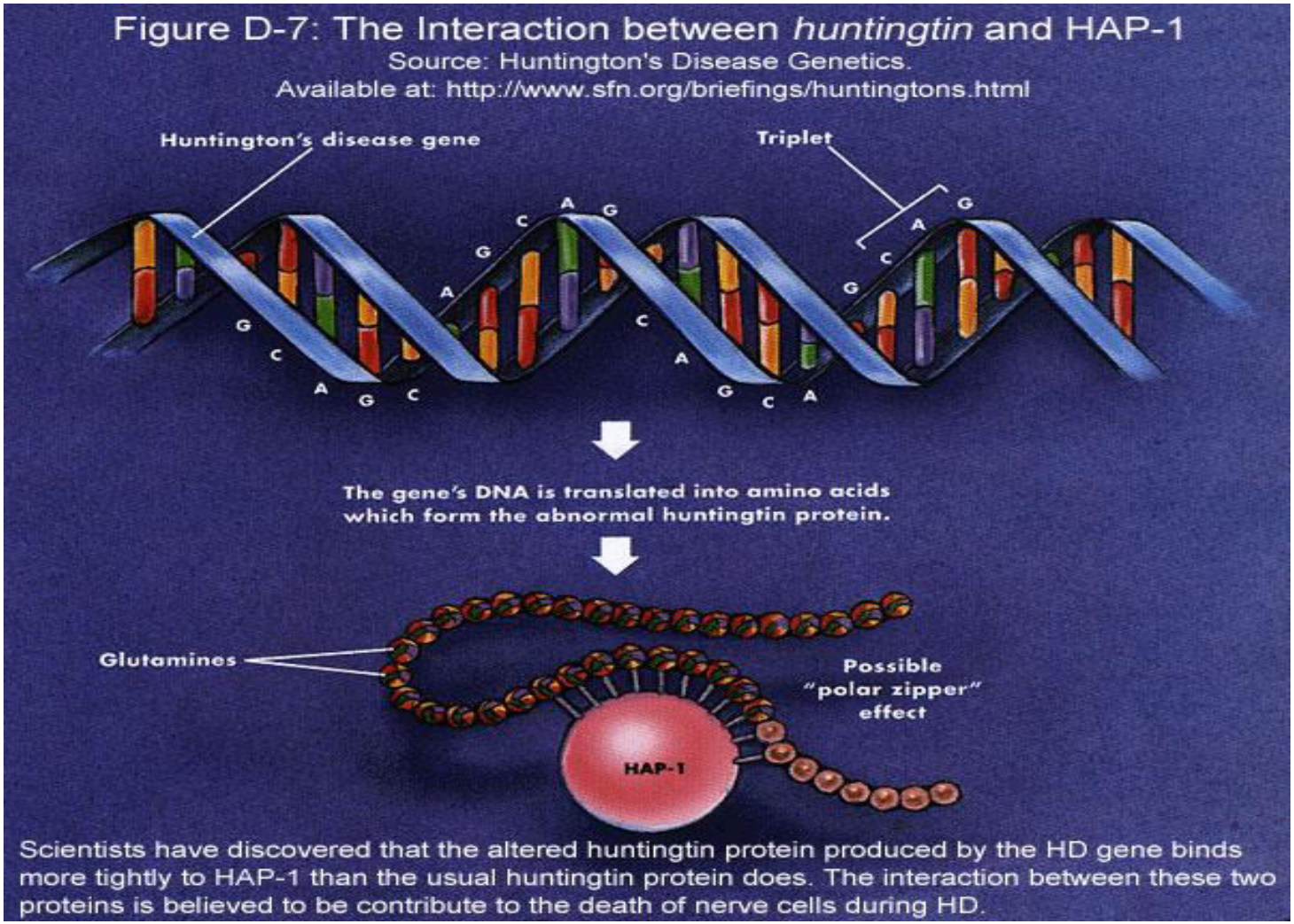
Autosomal dominant trait



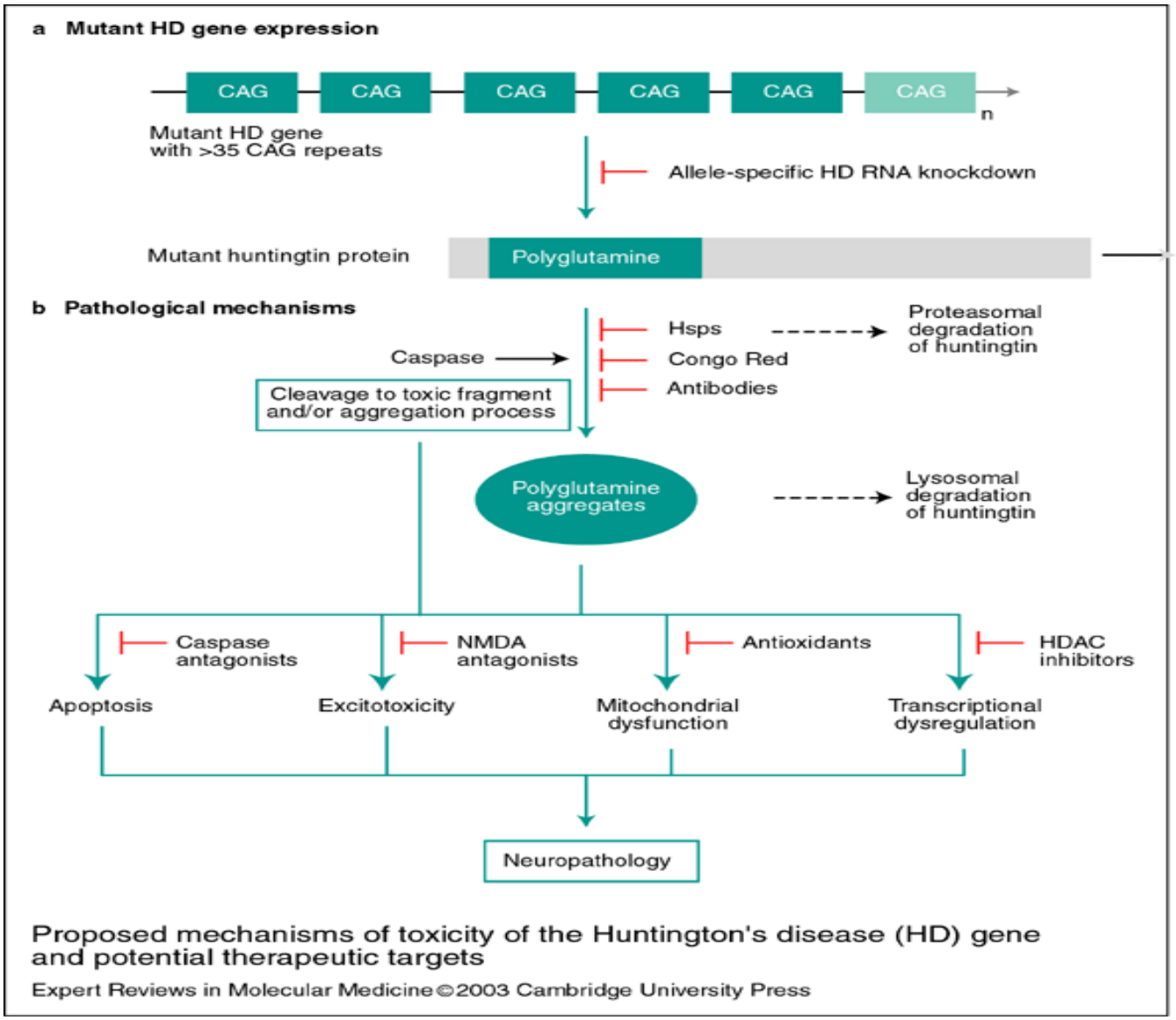
Autosomal dominant pedigree



Huntington's disease



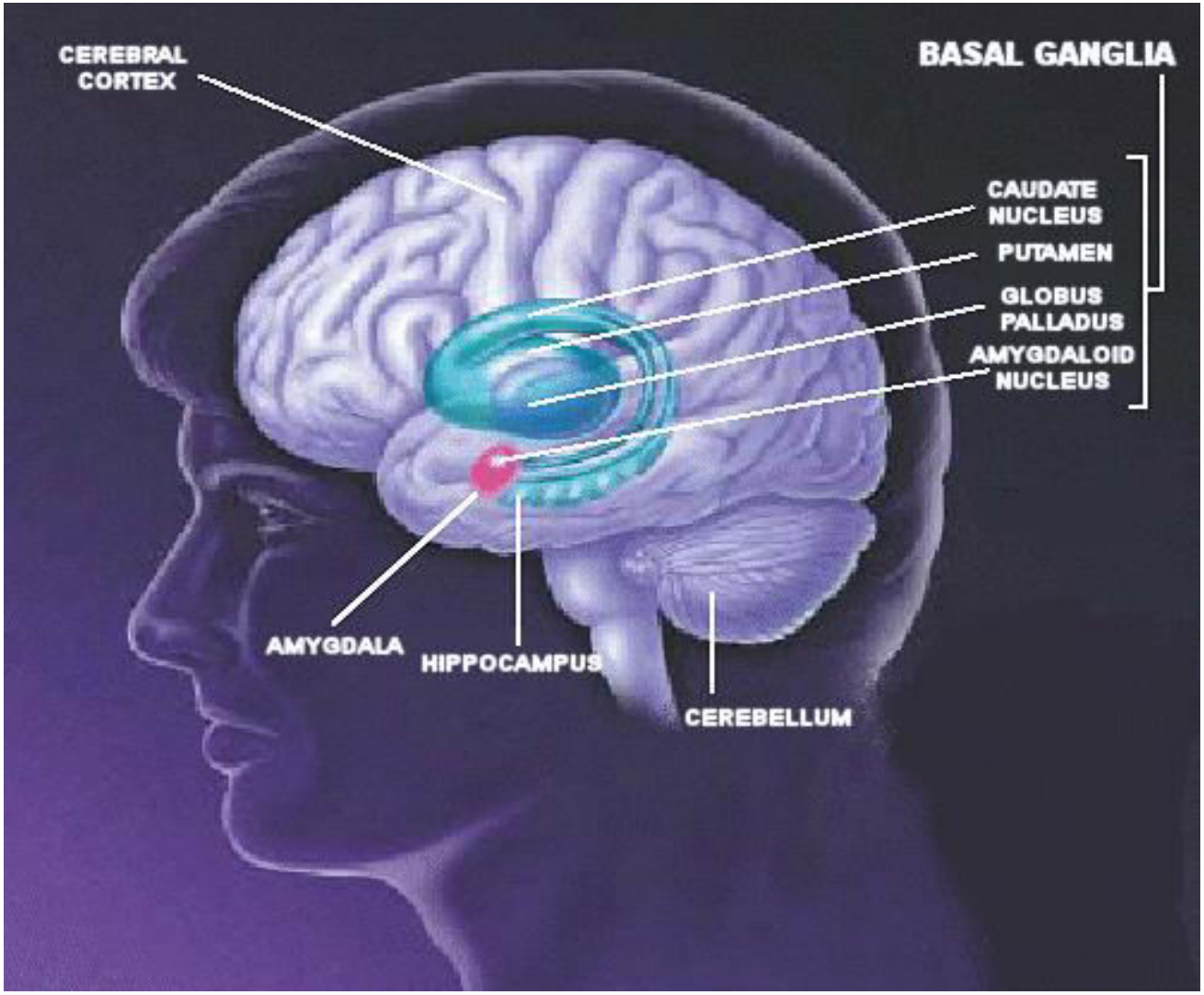
Huntington's disease



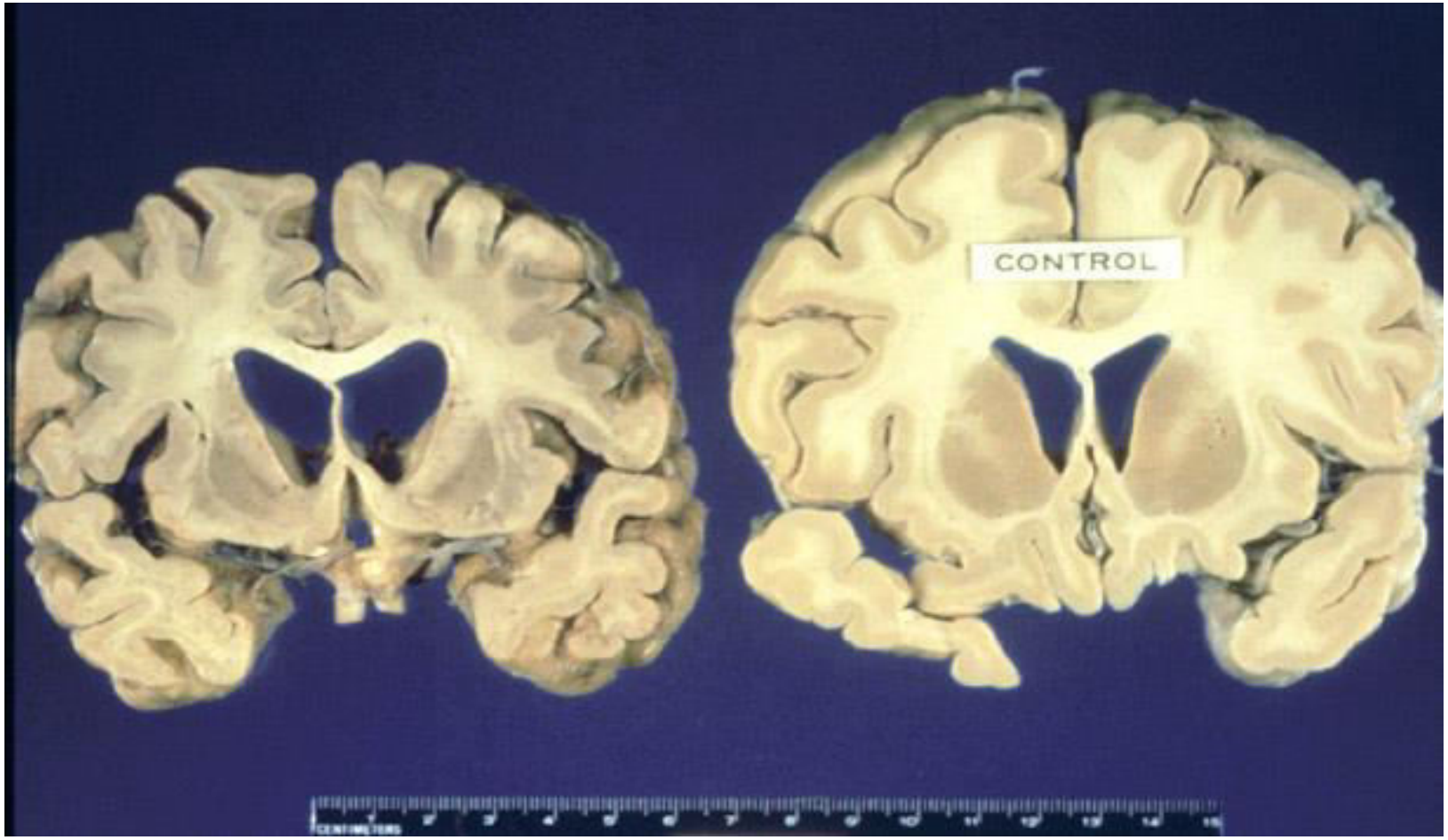
Proposed mechanisms of toxicity of the Huntington's disease (HD) gene and potential therapeutic targets

Expert Reviews in Molecular Medicine ©2003 Cambridge University Press

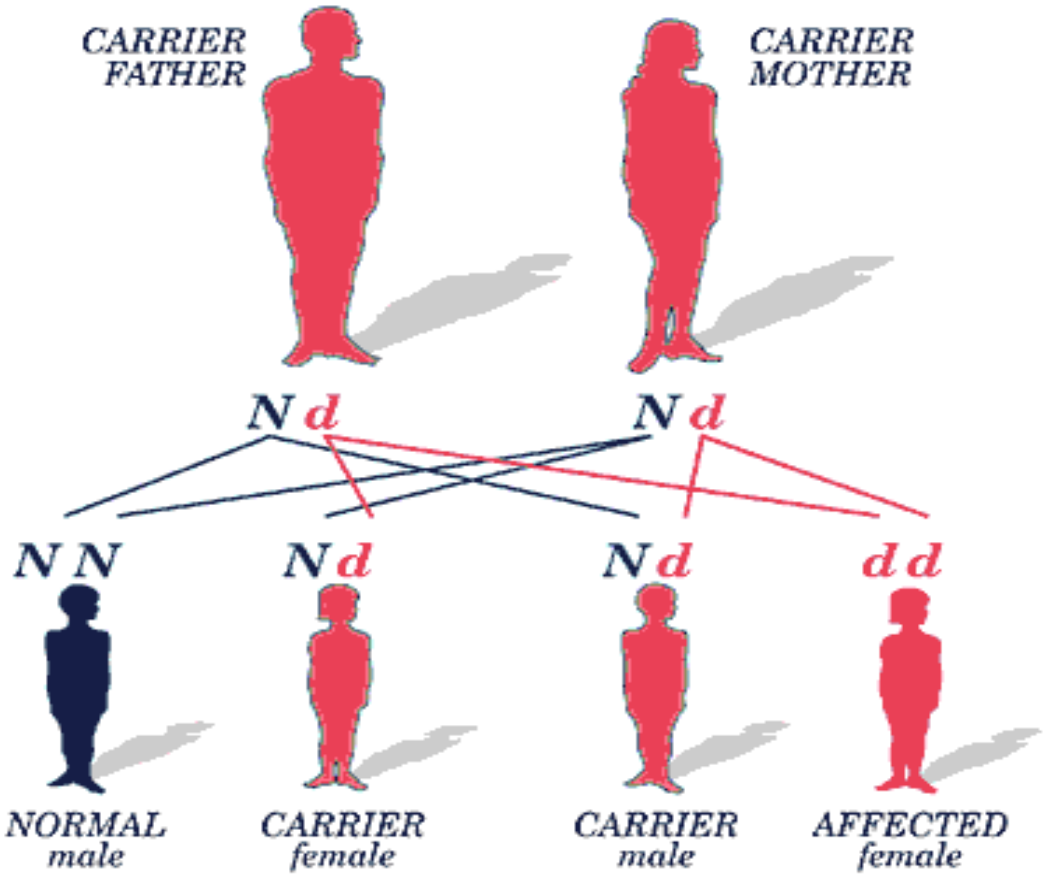
Huntington's disease



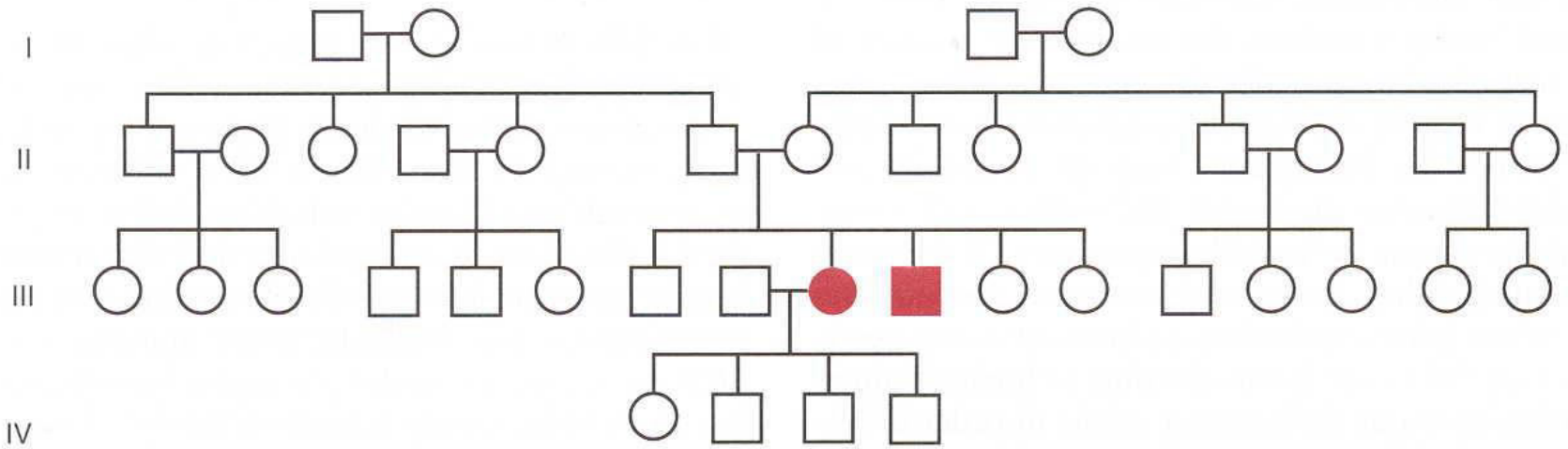
Huntington’s disease



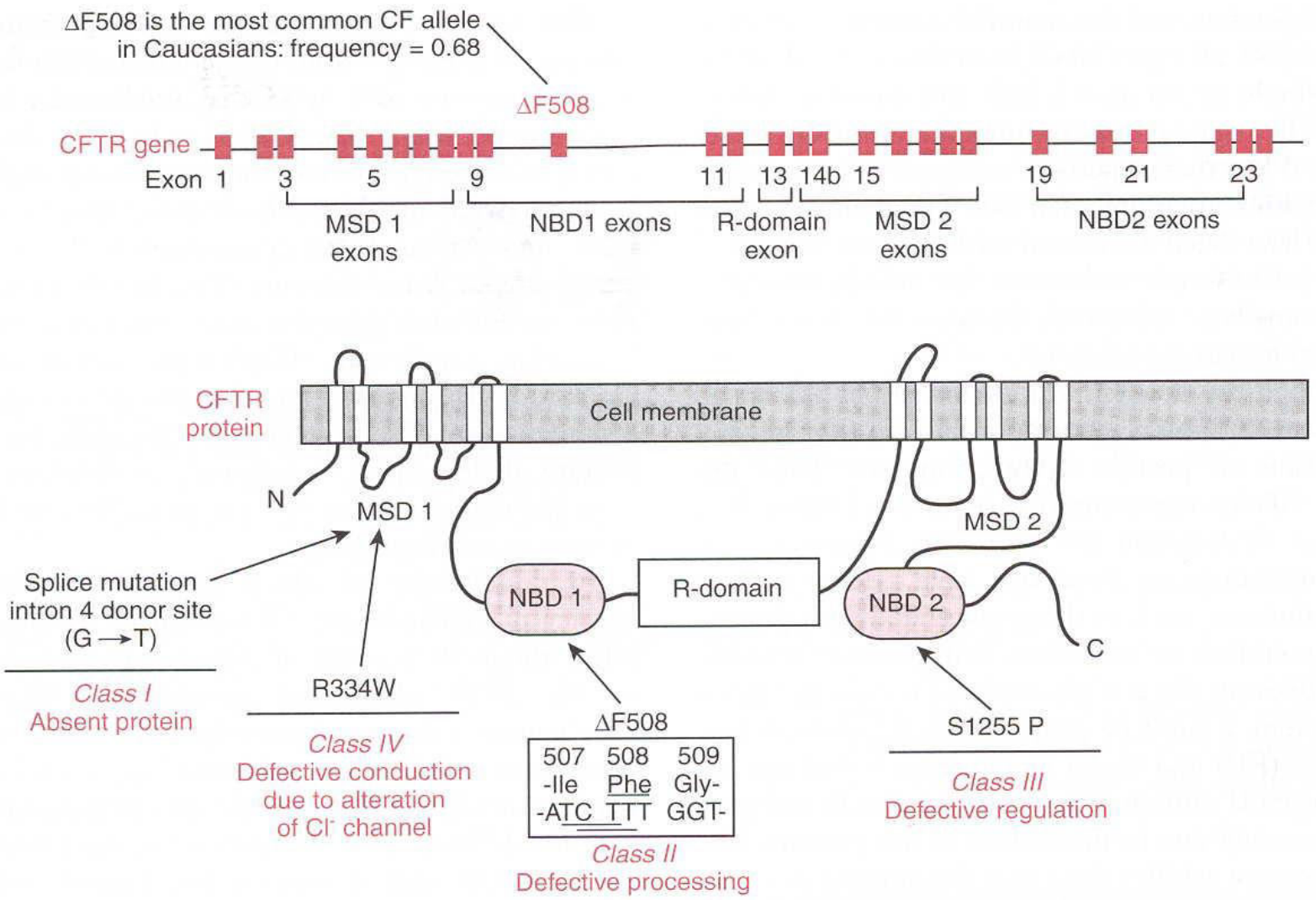
Autosomal recessive trait



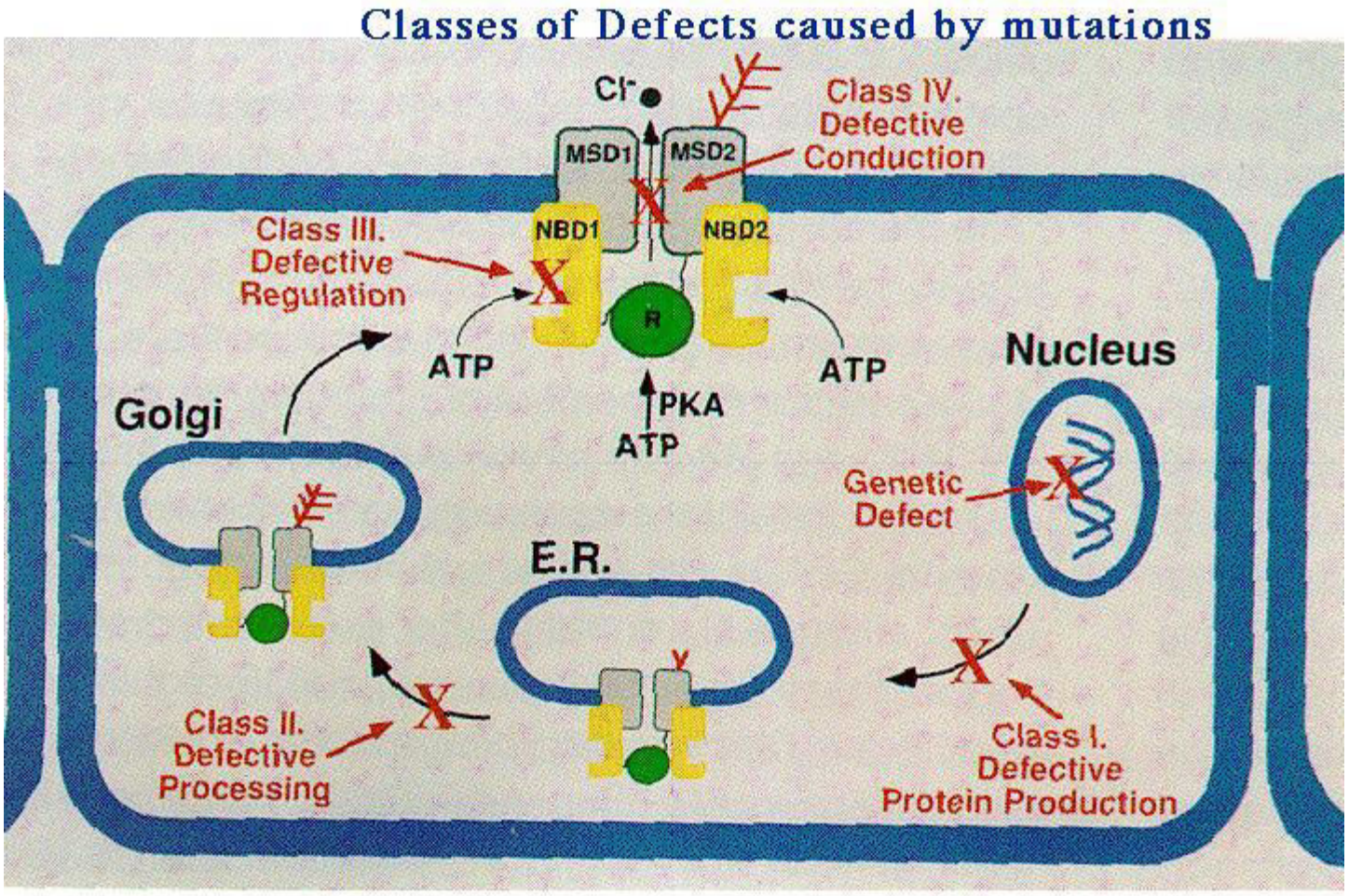
Autosomal recessive pedigree



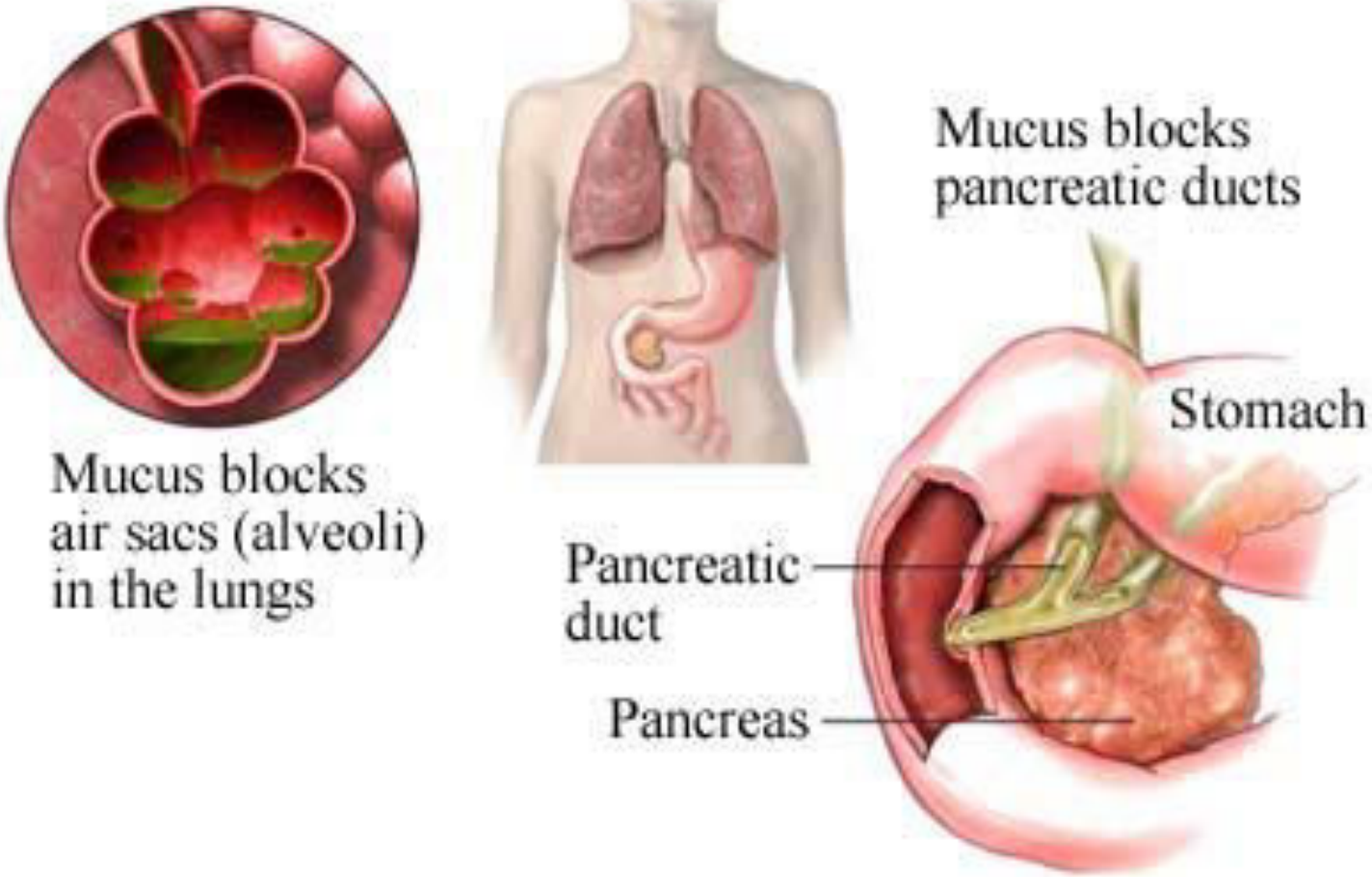
Cystic fibrosis

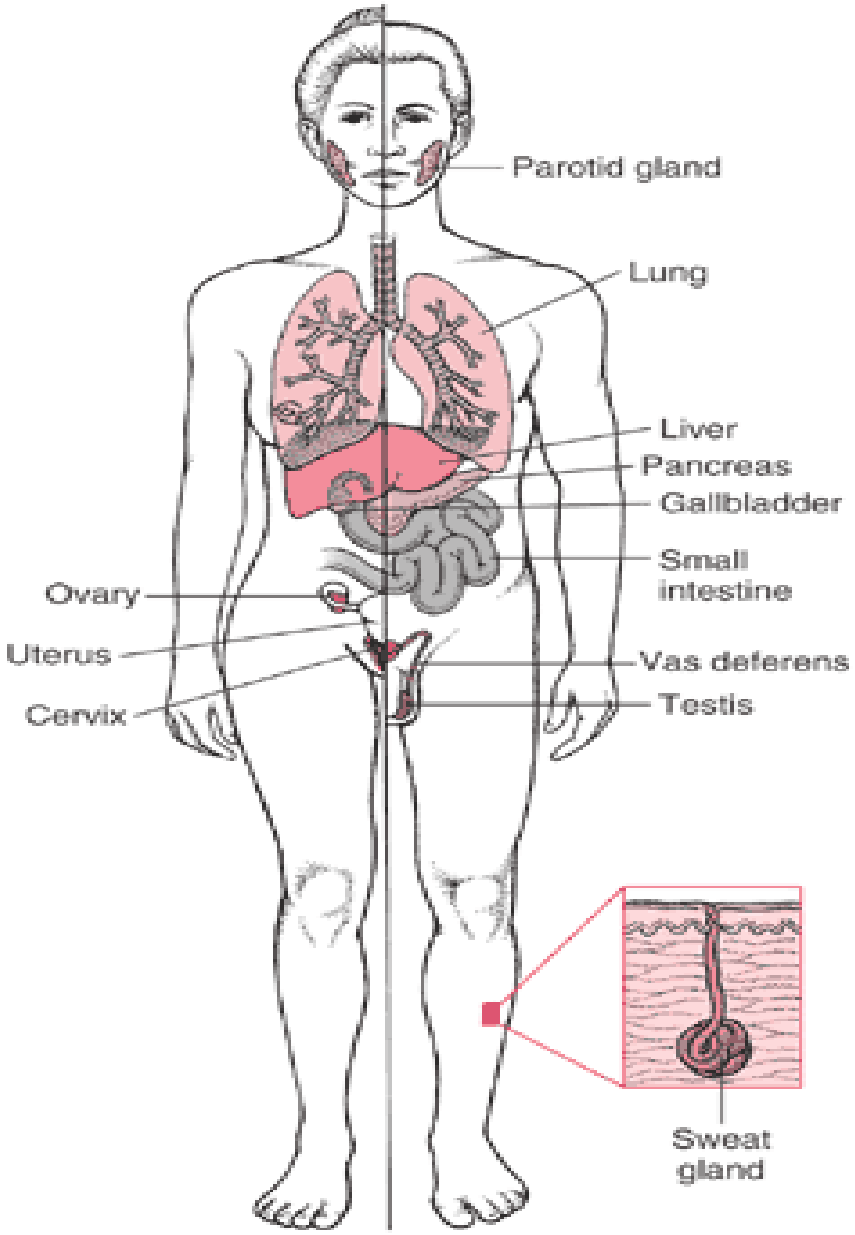


Cystic fibrosis



Cystic fibrosis

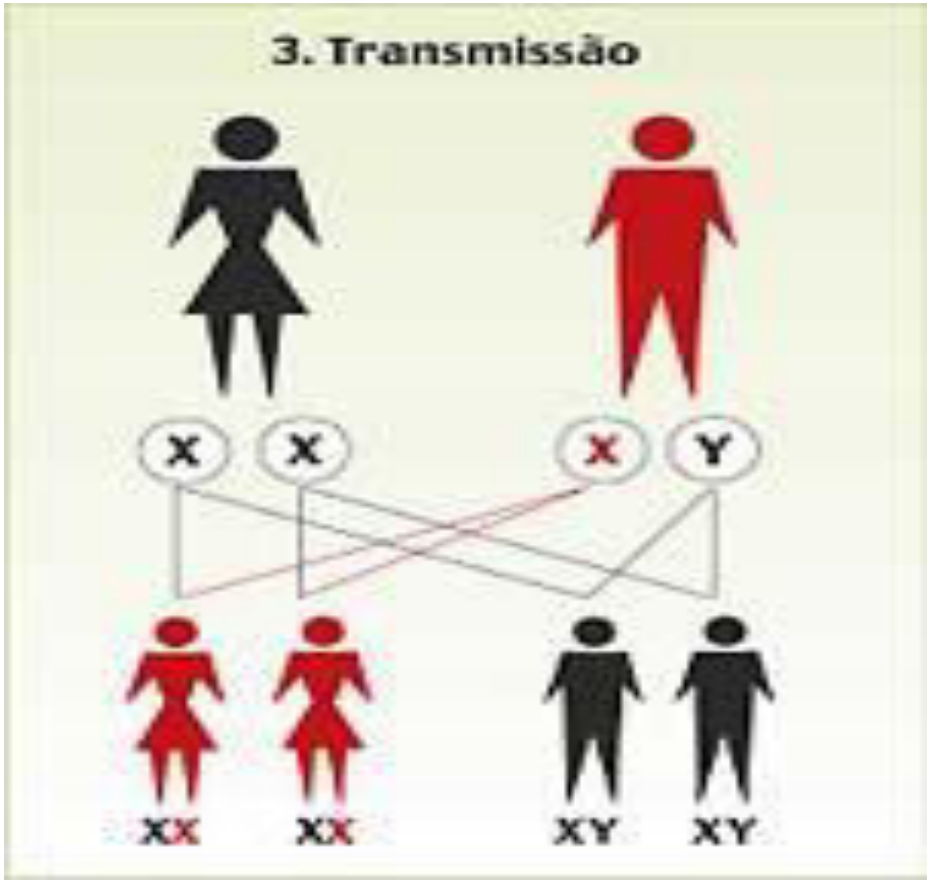




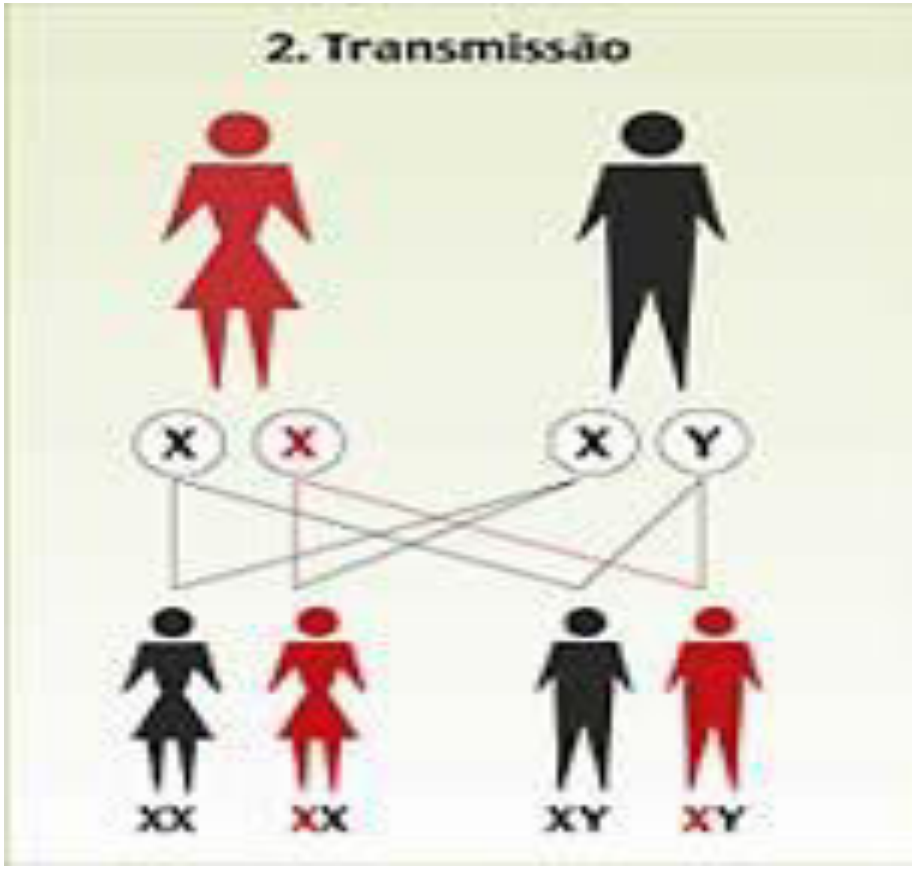
Cystic Fibrosis: Not Just a Lung Disease

In the lungs, thick bronchial secretions block the small airways, which become inflamed. As the disease progresses, the bronchial walls thicken, the airways fill with infected secretions, areas of the lung contract, and lymph nodes enlarge. In the liver, thick secretions block the bile ducts. Obstruction may also occur in the gallbladder. In the pancreas, thick secretions may block the gland completely. In the small intestine, intestinal obstruction (meconium ileus) can result from thick secretions and requires surgery in some newborns. The reproductive organs are affected by cystic fibrosis in various ways, often resulting in infertility. The sweat glands in the skin and small salivary glands in the cheek (parotid glands) secrete fluids containing more salt than normal.

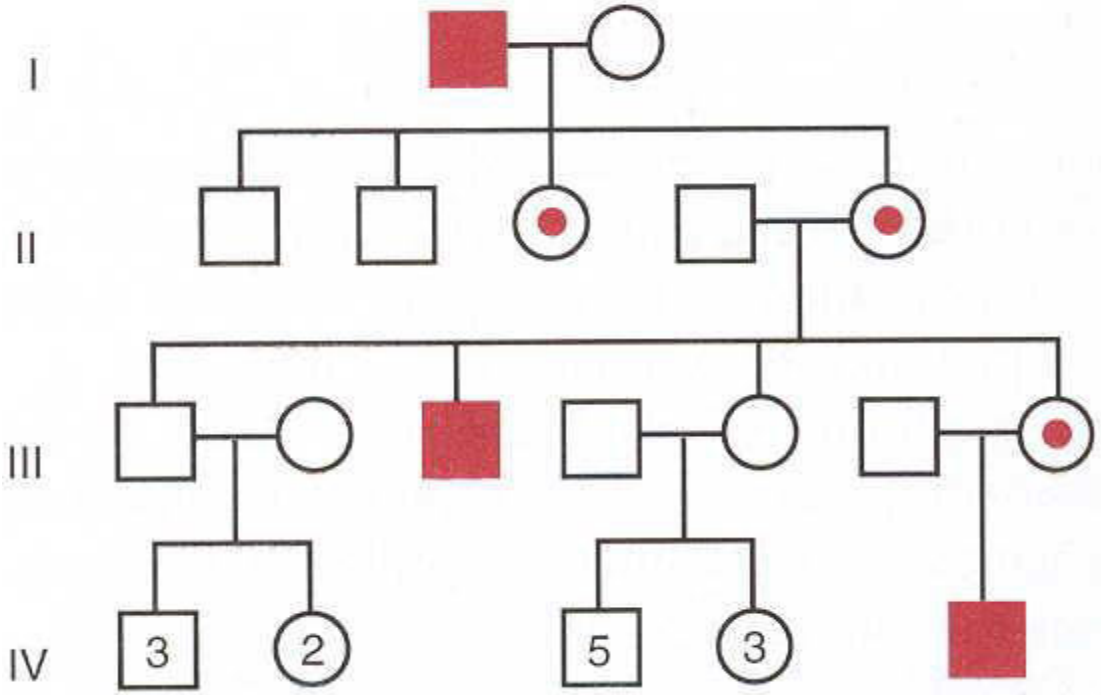
X-linked recessive trait



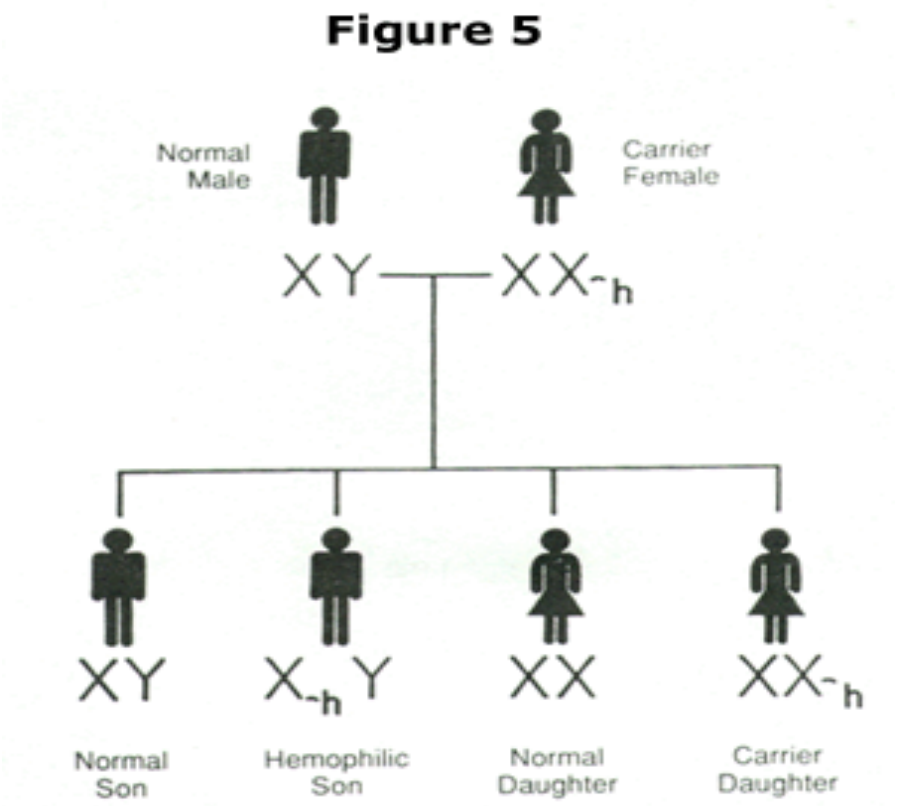
X-linked recessive trait



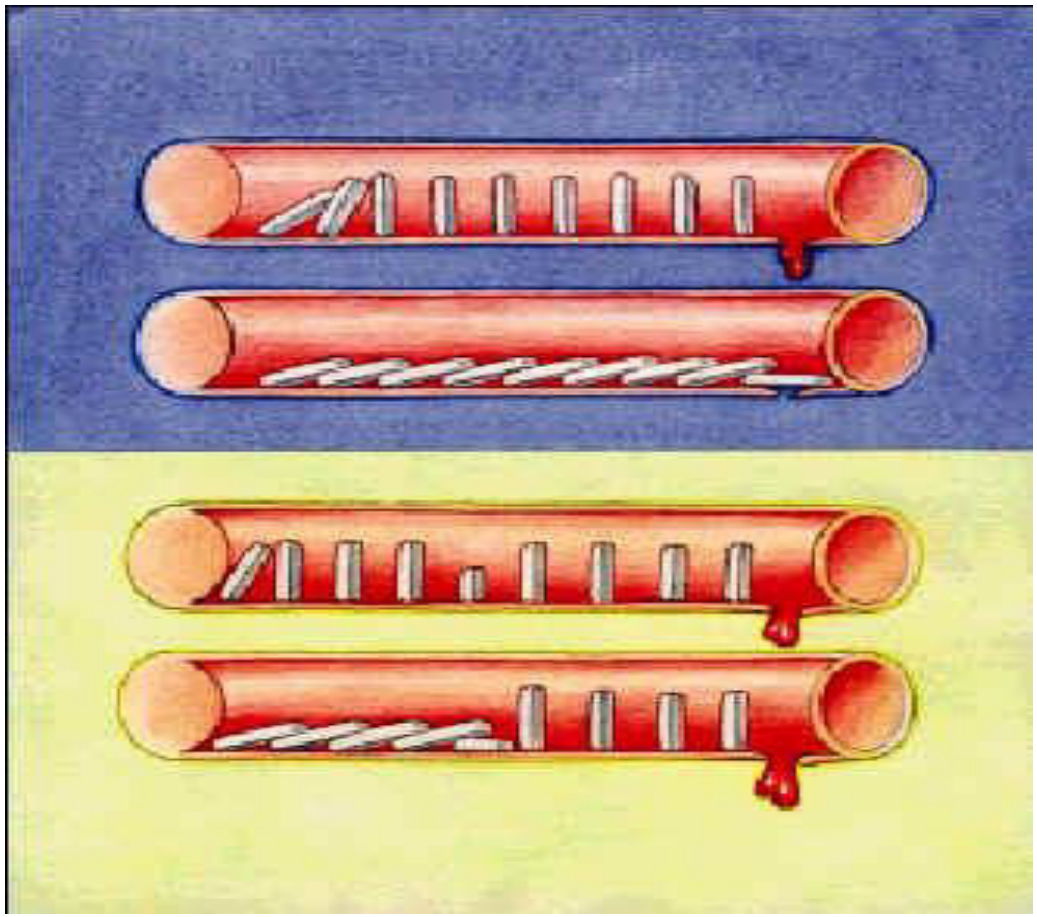
X-linked recessive pedigree



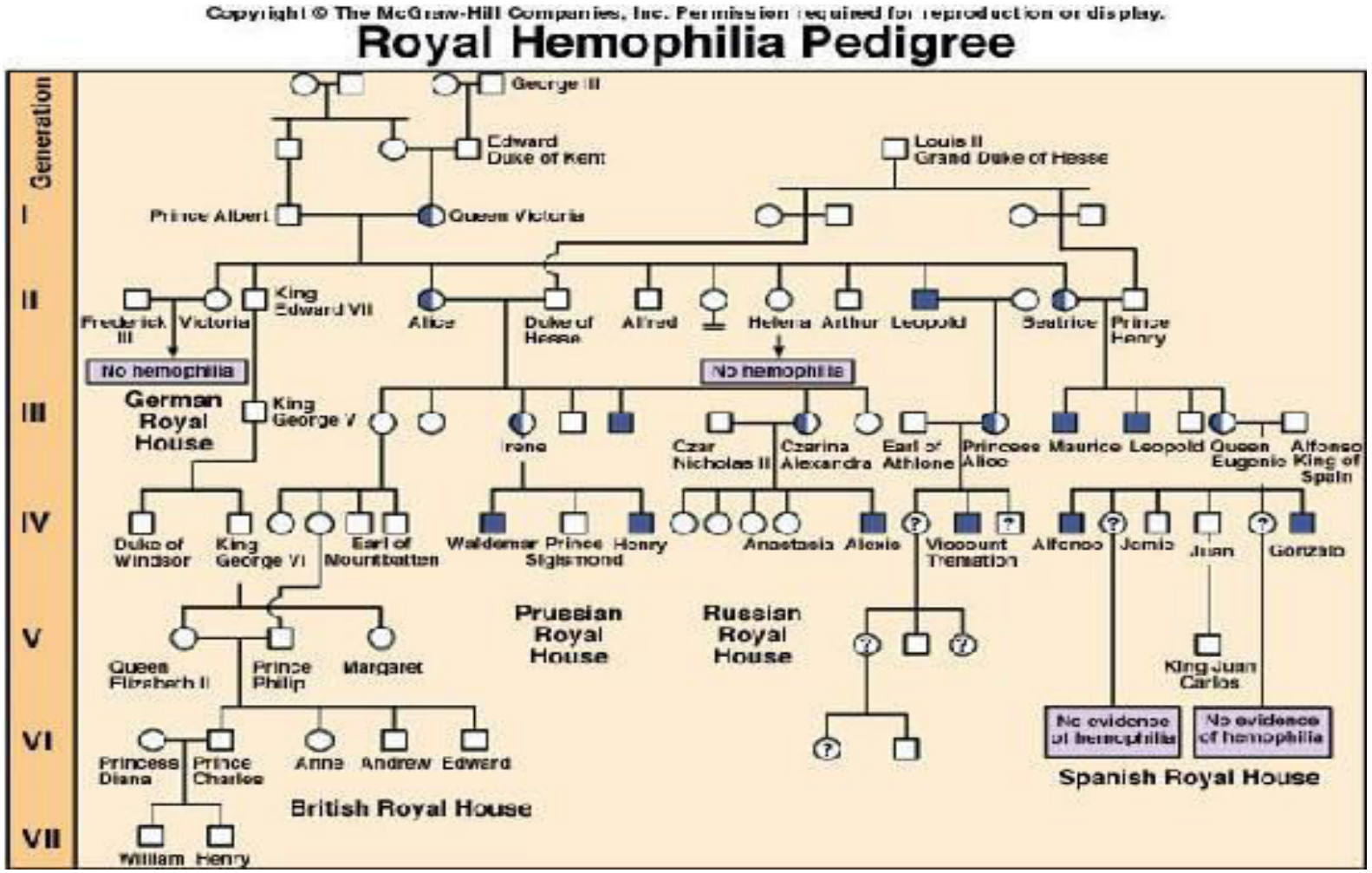
Hemophilia


Figure 5
Normal Male
Carrier Female
XY
XX-h
XY
X-hY
XX
XX-h
Normal Son
Hemophilic Son
Normal Daughter
Carrier Daughter

Hemophilia



Hemophilia


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Royal Hemophilia Pedigree
Generation
I
II
III
IV
V
VI
VII
George III
Edward Duke of Kent
Louis II Grand Duke of Hesse
Prince Albert
Queen Victoria
King Edward VII
Alice
Duke of Hesse
Alfred
Helena
Arthur
Leopold
Beatrice
Prince Henry
No hemophilia
No hemophilia
German Royal House
King George V
Irene
Czar Nicholas II
Czarina Alexandra
Earl of Athlone
Princess Alice
Maurice
Leopold
Queen Eugenie
Alfonso King of Spain
Duke of Windsor
King George VI
Earl of Mountbatten
Waldemar
Prince Sigismond
Henry
Anastasia
Alexis
Viscount Trematon
Alfonso
Jaime
Juan
Gonzalo
Queen Elizabeth II
Prince Philip
Margaret
Prussian Royal House
Russian Royal House
King Juan Carlos
Princess Diana
Prince Charles
Anne
Andrew
Edward
No evidence of hemophilia
No evidence of hemophilia
British Royal House
Spanish Royal House
William
Henry